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Face De-Identification Using Diffusion Models

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Deidentifikacija obrazov z difuzijskimi modeli

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Manfred Gonzalez Hernandez, 2025

To God and those who walked with me through darkness and light— your love, prayers, and grace carried me through the struggle.

"Through many dangers, toils and snares, I have already come; 'Tis grace has brought me safe thus far and grace will lead me home"

— John Newton

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List of used acronyms

acronym	meaning
EX	Expression Accuracy
GD	Gender Accuracy
AUC	Area Under the Curve
F1	F1 Score
FD	Fréchet Distance
MSE	Mean Squared Error
LFW	Labeled Faces in the Wild
CALFW	Cross-Age Labeled Faces in the Wild
RaFD	Radboud Faces Database
XM2VTS	The Extended M2VTS database
LoG	Laplacian of Gaussian
KD	Kernel Distance
GDDiff	GauDeiDiff
Acc	Accuracy
MAE	Mean Absolute Error
P	Precision
R	Recall
ROC	Receiver Operating Characteristic
TPR	True Positive Rate
FPR	False Positive Rate
FAR	False Acceptance Rate
FRR	False Rejection Rate
EER	Equal Error Rate
FP	False Positive
TP	True Positive
TN	True Negative
FN	False Negative

Abstract

Title: Face De-Identification Using Diffusion Models

In an era of increasing surveillance and data sharing, protecting individuals' identities in visual data has become a critical concern. This thesis addresses the problem of face de-identification using diffusion models, a powerful class of generative models capable of synthesizing high-fidelity images. We propose two novel approaches that modify the denoising trajectory of diffusion models through two different identity-independent and identity-dependent guidance regimes to generate de-identified facial images while preserving essential attributes such as expression and gender. Our method leverages pre-trained identity recognition models and preconditioned guidance to balance visual realism/quality and privacy. We evaluate our approach on four benchmark datasets—RaFD, XM2VTS, LFW, and CALFW—using metrics such as AUC, F1 Score, Fréchet Distance, and Mean Squared Error. The results demonstrate comparable performance with state-of-the-art techniques in both identity removal effectiveness and attribute preservation. Additionally, we analyze the behavior of the diffusion process under different guidance strategies, providing insight into the trade-offs between identity concealment and image quality.

Keywords

Face de-identification, diffusion models, generative models, privacy preservation, preconditioned guidance, identity recognition, attribute preservation

Povzetek

Naslov: Deidentifikacija obrazov z difuzijskimi modeli

V dobi vse večjega nadzora in deljenja podatkov postaja zaščita identitete posameznikov na vizualnih podatkih ključnega pomena. Magistrsko delo obravnava problem anonimizacije obrazov z uporabo difuzijskih modelov—zmogljivega razreda generativnih modelov, sposobnih sinteze visokokakovostnih slik. Predlagamo dva nova pristopa, ki spreminjata pot razšumljanja difuzijskih modelov preko dveh različnih režimov vodenja: enega neodvisnega od identitete in drugega odvisnega od identitete. Cilj je ustvariti anonimizirane slike obrazov, pri tem pa ohraniti bistvene značilnosti, kot sta izraz in spol. Naš pristop uporablja vnaprej naučene modele za prepoznavo identitete ter predpogojeno vodenje, s čimer dosežemo ravnovesje med vizualnim realizmom in zasebnostjo. Naš pristop ocenjujemo na štirih referenčnih zbirkah podatkov—RaFD, XM2VTS, LFW in CALFW—z uporabo metrik, kot so AUC, F1 rezultat, Fréchetova razdalja in srednja kvadratna napaka (MSE). Rezultati kažejo primerljivo učinkovitost z obstoječimi vrhunskimi tehnikami tako pri odstranitvi identitete kot pri ohranjanju atributov. Poleg tega analiziramo vedenje difuzijskega procesa pri različnih strategijah vodenja in s tem osvetlimo kompromise med prikrivanjem identitete in kakovostjo slike.

Ključne besede

Anonimizacija obrazov, difuzijski modeli, generativni modeli, varovanje zasebnosti, predpogojeno vodenje, prepoznavanje identitete, ohranjanje atributov

Chapter 1

Introduction

Privacy in facial images is increasingly crucial. It is essential to safeguard individuals' data and control access to sensitive information embedded in these images. This aligns with privacy definitions [1], which refer to the right to be left alone, freedom from interference or intrusion, and control over how personal information is collected and used. Privacy is recognized as a fundamental human right, specially protected under Act 1998, Article 8: Right to Privacy [2]. Various commercial software solutions address the need for user privacy protection, for example Brighter AI[1] offers a solution that either blurs the subject's face or applies a generative technique to create a synthetic overlay, effectively safeguarding individuals from recognition. Similarly, applications like Gallio[2] use blurring techniques to preserve privacy in images and videos. However, naive approaches like blurring [3] alone may be vulnerable to imitation attacks and parrot attacks [4, 5], exposing these applications to potential risks [6].

On the other hand, data utility in facial images remains a major concern, as high visual quality often correlates with identifiable features. Early approaches like blurring [3], down-sampling [7], occlusions [8], pixelation [9] or deliberate masking [4] often compromise data utility by significantly de-

[1] https://brighter.ai/
[2] https://gallio.pro/

grading facial information. More recent deep-learning–based methods exploit generative neural networks to synthesize facial images; however, these outputs often appear artificial and lack photorealism [10, 11, 12, 13]. Additionally, methods based on autoencoders [14, 15, 16] and generative adversarial networks (GANs) [17, 18, 19, 20, 21] face challenges in balancing natural appearance with effective deidentification, often failing to prevent re-identification by state-of-the-art (SOTA) recognition models.

In recent years, a novel image generation approach called Diffusion Models (DMs) has emerged and gained popularity alongside GANs. Diffusion Models generate images that approximate actual images with impressive levels of resolution, quality and diversity [22]. Furthermore, representative approaches such as Latent Diffusion [23] and Conditional Diffusion [24] models have been developed within this framework. Building on these advancements, an innovative approach was introduced [25] for image restoration using diffusion models with iterative preconditioned guidance, also known as DDPG. The proposed work highlights the potential of back-projection (BP) [26] and least squares (LS) guidance mechanisms to enhance the inverse process, achieving near-perfect, high-quality facial image reconstructions.

On the other hand, as AI models grow in scale, so does their energy consumption and carbon footprint, raising concerns about sustainability. Bender et al. [27] highlight that large-scale models exacerbate CO_2 emissions, disproportionately impacting marginalized communities who bear the environmental costs without equitable access to AI's benefits. This calls for a paradigm shift toward energy-efficient AI architectures.

To address these issues, we introduce **MaskDDPG** and **GauDeiDiff**, novel frameworks built on DDPG that enable high-quality deidentified facial image generation without any additional training and set a new state-of-the-art among diffusion-based deidentification models. Central to our MaskDDPG approach is a controllable privacy level, which adjusts the strength of guidance in the diffusion Markov chain; in combination with a segmentation mask, this privacy level lets us specify which facial regions to anonymize.

As this privacy level increases, the generated image progressively diverges from the original identity—removing identifying information while preserving non-identity features—and, when the privacy level approaches values near 1, the model can synthesize fully out-of-distribution identities, achieving the highest levels of anonymity and enhanced privacy protection.

Additionally, our second approach, GauDeiDiff, built on DDPG, generates high-quality deidentified facial images while ensuring privacy protection without having to train or fine-tune the diffusion model again. Our solution applies a Gaussian kernel and a Laplacian of Gaussian Kernel over the back-projection (BP) and least-squares (LS) guidance tensors specifically for face deidentification. This application integrates, in an incremental manner, original facial features from the input image with the learned distribution from training.

In our experiments, we apply the AdaFace face recognition model [28] to publicly available benchmark data sets to evaluate the privacy protection achieved by our mechanism. We then assess data utility by quantifying how well key facial features are retained using standard utility metrics, and evaluate data quality by measuring image-resolution preservation.

This thesis comprises seven chapters. After the Introduction (Chapter 1), Chapter 2 surveys current methods for synthetic image generation and related work. Chapter 3 provides a theoretical overview of denoising diffusion models with iteratively preconditioned guidance. In Chapter 4, we present **MaskDDPG**, our novel framework for producing de-identified facial images while preserving non-identity features. Chapter 5 introduces **GauDeiDiff**, in which we leverage kernel techniques to generate highly private synthetic samples. Chapter 6 describes the facial datasets and evaluation metrics used in our experiments, and compares our methods against state-of-the-art models both qualitatively and quantitatively, with a detailed analysis of the results. Finally, Chapter 7 summarizes our key findings, discusses the limitations of our approaches, and outlines directions for future research.

Chapter 2

Related Work

In this chapter, we explore existing research work related to the topic of our thesis. In the following sections, we provide an overview of recent advancements in deep generative methods and cover approaches specifically designed for face generation and face deidentification. Lastly, we present various diffusion models solutions used for the automatic creation of deidentified datasets.

2.1 Deep Generative based methods

Recent advances in face deidentification balance privacy preservation and image quality by leveraging deep generative models, including autoencoders, various encoder–decoder architectures, and GAN-based models. One-of-the-earliest deep generative approaches proposed by Gong et al. [29], with a dual-VAE (Variational Autoencoder) architecture to disentangle identity-related features, achieving privacy without sacrificing utility. To this end, dual-VAE utilizes a chained representation-learning framework using two parameter-sharing VAEs to separate identity-related and identity-independent factors, ensuring targeted obfuscation of identity without perturbing other facial attributes. To fully preserve facial semantics, a factor-invariant strategy with combinatorial constraints resulted in disentangled attributes without any

auxiliary labels. A deidentified-face-guided inpainting network then restores fine visual details, achieving a balance between privacy and image quality. Similarly, UU-Net [30], a reversible approach, enables secure deidentification and controlled identity recovery in surveillance footage. UU-Net introduces a "double-U" encoder–decoder that first anonymizes faces under the guidance of a pairwise attribute matcher and adversarial losses, then reversibly reconstructs the originals. At test time, it crops and passes head regions through the encoder for de-identification, hides these regions via steganography in the output stream, and allows the decoder—available only to authorized parties—to recover the true faces.

GAN-based approaches have advanced significantly, exemplified by CPP-DeID [31], which incorporates both the face and surrounding context through GAN inversion, latent optimization, and a multi-objective loss to offer a trade-off between data utility preservation and privacy protection. Other notable GAN models include CIAGAN [17], which ensures temporal consistency in video anonymization by leveraging a conditional GAN to generate realistic anonymized images and utilizes an identity-guidance discriminator to control identifying characteristics. On the other hand, RIDDLE [32] employs cipher-guided encryption for reversible anonymization, where each training step consists of two forward propagations: one for encryption and another for decryption.

Further advancements include Semi-Adversarial Networks [14], which employ a convolutional autoencoder to perturb face images, a convolutional neural network (CNN) as an auxiliary face matcher, and a CNN as an auxiliary gender classifier for flexible soft-biometric privacy preservation. Evolutionary GAN [33] maintains structural similarity in deidentified faces by modifying the generator's loss function to incorporate distance and structural similarity measures between the original and deidentified faces. AMT-GAN [21] employs adversarial makeup transfer from reference images and a regularization module, along with a joint training strategy, to reconcile conflicts between adversarial perturbations and cycle-consistency loss in makeup transfer.

The CLEANIR model [16] employs VAEs to modify identity information, the authors present an end-to-end de-identification framework based on a variational auto-encoder (VAE) that explicitly disentangles identity information from other facial attributes. By decomposing the latent representation into an identity vector and an attribute vector, the method modifies only the identity component—thereby preserving pose, expression, age, and other non-identity cues—while allowing fine-grained control over the degree of anonymization via simple transformations in latent space. Compared to GAN-based approaches, which often suffer from convergence issues and artifact generation, the VAE backbone yields more coherent and natural outputs. QM-VAE [34] introduce the Quality Maintenance–Variational Autoencoder (QM-VAE), a novel VQ-VAE-based framework that first anonymizes facial images via multiple established protection methods and then fuses them under a service-oriented loss to preserve downstream utility. By integrating vector quantization into the VAE bottleneck, QM-VAE produces high-fidelity, de-identified faces and, through a black-box service-quality evaluator plugged directly into the training objective, automatically balances privacy with task-relevant signal.

Action-preserving anonymization was proposed by Ren et al. [35] to tackle the challenge of anonymizing faces in video streams so as to protect individual privacy without degrading downstream action-detection performance. Ren et al. [35] formulate an adversarial, multi-task learning framework in which a face modifier network is trained to perturb each detected face so that a face classifier (the adversary) can no longer recognize the identity, while simultaneously an action detector is optimized to correctly localize and label actions in the modified frames. To ensure visual coherence, they also include an L_1 "photorealism" loss so that poses, expressions, and scene context remain intact. DeepPrivacy (DP) [36] and a second-generation DeepPrivacy2 (DP2) [20] offer advanced style-based U-Net GAN and a multi-modal detection/stitching pipeline. Their system first detects every person via an ensemble of DensePose (CSE), Mask-R-CNN, and DSFD face detectors, then

applies one of three inpainting generators—(1) CSE-guided full-body, (2) unconditional full-body, or (3) face GAN—to synthesize high-fidelity, diverse replacements that cover hair and accessories. A recursive, area-aware stitching orders overlapping outputs for seamless composition. They also enable text-driven attribute editing via latent-space manipulations.

2.2 Diffusion Models

Diffusion Models (DMs) have the potential of significantly advancing face generation approaches, with Collaborative Diffusion [37] achieving deidentification without retraining. Collaborative Diffusion represents a significant advance in multi-modal image synthesis by demonstrating how independently pre-trained, uni-modal diffusion models can be effectively coupled to perform complex face generation and editing tasks without any expensive retraining. At its core, the framework introduces lateral "collaborative" connections between models—such as a text-driven age predictor and a mask-driven shape modeller—so that, during each denoising iteration, a light-weight "dynamic diffuser" meta-network learns spatial–temporal influence functions that adaptively weight and fuse their outputs. This design not only preserves the flexibility to mix and match off-the-shelf architectures and modalities, but also affords fine-grained, stage-wise control over each condition's contribution.

DCFace [38] introduces a compelling two-stage paradigm for large-scale synthetic face dataset generation that directly addresses the key challenges of subject uniqueness, style diversity, and label consistency. In the first "condition sampling" stage, a high-quality identity image of a new subject is synthesized, and an arbitrary real-data style image is drawn from a style bank; in the subsequent "mixing" stage, these two conditions are fused by Gmix—a dual-condition denoising diffusion probabilistic model—that produces images embodying the identity of the new input subject with the visual style of the real-data style drawn from the bank. To rigorously enforce label consistency, the authors introduce a time-step dependent ID loss that

interpolates identity features over the diffusion trajectory, ensuring that the generated outputs remain faithfully tied to their source identities. Beyond its core generative engine, DCFace also incorporates a patch-wise style extractor to preserve spatial detail.

In the field of face de-identification using diffusion models, FaceDiffusion [39] balances efficiency and realism for resource-constrained embedded systems. FaceDiffusion introduces a lightweight, two-stage anonymization pipeline tailored for resource-constrained "TinyML" platforms that balances user privacy with the preservation of essential facial structure. In the first stage, a Denoising Diffusion Probabilistic Model (DDPM [40]) synthesizes low-resolution (64×64) anonymized faces; in the second stage, super-resolution networks upscale these outputs to 256×256, yielding visually coherent images while maintaining keypoint fidelity. CAMOUFLaGE [41] pioneers the use of Latent Diffusion Models (LDMs [23]) for holistic image anonymization by transforming entire scenes—faces, backgrounds, and context—rather than simply obscuring facial regions. It offers two variants: CAMOUFLaGE-Base, which harnesses an ensemble of pretrained ControlNets (depth, normals, segmentation, pose, and edges) alongside a novel "negative control" to steer generated outputs away from their originals, and CAMOUFLaGE-Light, a streamlined framework that conditions generation on image embeddings and captions via an IP-Adapter [42] and preserves individual facial features through a T2I-Adapter [43].

LDFA [44] presents a targeted two-stage pipeline for face anonymization in Intelligent Transportation Systems that prioritizes both privacy and the integrity of downstream perception tasks. In its first phase, high-precision face regions are localized via RetinaFace while the second phase replaces each detected face with a lifelike in-painting generated by a pre-trained LDM, requiring no additional fine-tuning. Although LDFA faces challenges with small faces, artifacts, and overlapping regions. Diff-Privacy [45] introduces an energy-function–based identity guidance module, which injects gradient corrections during the denoising trajectory to steer the model toward ei-

ther correct or intentionally incorrect identity recognition by downstream classifiers—and, by varying the random noise inputs, maximizes inter-image identity diversity for stronger anonymization. Recognizing that different diffusion timesteps capture different semantic scales, the authors further propose a multi-scale image inversion module coupled with adaptive embedding-scheduling strategies, allowing the system to tailor the degree of privacy protection to specific requirements.

In the other hand, DIFP [46] provide a diffusion-based framework that encrypts facial identities with photorealistic, high-resolution surrogates while preserving background, pose, and stylistic attributes—and yet allows near-lossless recovery by authorized parties. At its core, DIFP approach is dual-conditioned on both identity and style embeddings, ensuring that encrypted outputs remain natural, diverse, and faithful to the original scene. A two-stage training regimen first co-trains on full identity/style signals to forge strong correlations, then fine-tunes with partial conditioning to boost generalization and the variety of encryptions per face. To further refine output realism, iterative latent-space adjustments governed by a "realism scale" are applied, yielding images that pass muster for downstream tasks like face detection. Finally, the original visage can be precisely restored via a targeted diffusion-model denoising process.

FAMS [47] employs a diffusion-based framework that replaces traditional landmark or mask-guided anonymization with a single reconstruction loss, yielding high-fidelity face swaps that obscure identity while preserving expressions, gaze, pose, and background context. Their model builds on a latent-diffusion UNet augmented by two lightweight "ReferenceNet [48]" modules—one encoding source (identity) features, the other encoding driving (non-identity) attributes—which are fused via cross- and self-attention to guide the synthesis process. The system is trained both conditionally (with a source image) and unconditionally, enabling anonymization from a single input by selectively blending conditional and unconditional model states.

FADM [49] propose a context-aware full-body anonymization pipeline

that leverages text-to-image diffusion models to replace every person instance in an image with a photorealistic surrogate while retaining critical scene semantics for downstream tasks. The method first detects and segments each person with instance masks, then in-paints those regions via a masked diffusion process guided by positive ("high quality, RAW photo, 8k") and negative ("cartoon, deformed") prompts and a noise-level parameter that smoothly interpolates between minimal and strong anonymization. To recombine multiple anonymized instances, FADM sorts crops by pixel-coverage and merges them back-to-front, ensuring occlusion handling. IDDiffuse [50] enables dual-conditional diffusion framework that anonymizes faces by first mapping each real identity to a consistent synthetic counterpart via an ensemble-trained identity mapper, and then fusing this synthetic identity with patch-wise texture features from the original image within a DDPM-based U-Net. A time-step-dependent anonymization loss smoothly interpolates between real and synthetic identity features while a texture-consistency term preserves non-identity details such as expression and pose.

DeDiM [51] employs a classifier-guided diffusion approach for face de-identification by integrating a pre-trained DDPM generator with a face-recognition model to steer sampled outputs away from the original identity while preserving visual fidelity. Rather than starting diffusion from pure noise, DeDiM initiates sampling at an early timestep—retaining most structural cues from the input—and then applies classifier guidance via identity-probability gradients from the recognition network to subtly perturb identity-bearing features. A binary landmark-based mask further blends original pixels around key facial regions (eyes, eyebrows, nose, and mouth) with the diffusion output, ensuring that non-identity details remain intact.

Finally, another video data focusing approach [52] extends anonymization to videos, this framework extends face de-identification from static images to videos by ensuring temporal coherence of anonymized identities. Their method employs a diffusion autoencoder to disentangle each frame's features into a unified, time-independent identity representation (captured by

an identity encoder) and temporally correlated motion features (extracted via a landmark encoder), while encoding background and non-identity details into low-dimensional random subcodes. During inference, this shared identity code is edited to obscure personal identity, and the conditional reverse DDIM process reconstructs each frame, yielding continuous, flicker-free video sequences. To prevent leakage of identity information into the background codes, the authors design an internal consistency loss (Linner) and an external consistency loss (Louter) alongside the standard DDPM reconstruction objective, which collectively enforce that only the semantic identity channel is altered while other attributes remain stable. These advancements highlight the ongoing progress in face deidentification, balancing privacy, image quality, and application-specific needs.

Although iterative preconditioned guidance has demonstrated great promise for high-quality facial image restoration [25], to our knowledge it has not yet been adapted for privacy-preserving deidentification. In the next chapters, we close this gap by leveraging back-projection and least-squares guidance within a diffusion framework—first in MaskDDPG, where a tunable guidance strength λ and segmentation masks enable region-specific anonymization, and then in GauDeiDiff, where Gaussian and Laplacian of Gaussian kernels refine the guidance tensors for even stronger anonymity without retraining the base model.

Chapter 3

Theoretical background

This chapter serves as an introduction to Diffusion Models. In it, we first go over the inner workings of the initial Denoising Diffusion Probabilistic Model (DDPM), then examine the evolution of Denoising Diffusion Implicit Model (DDIM) and dive deeper in Denoising Diffusion Models with Iteratively Preconditioned Guidance (DDPG), in order to enable a better understanding of the contributions presented in this thesis.

3.1 Denoising Diffusion Probabilistic Model

Diffusion Models are characterized by their image generation process based on step-wise denoising process. This approach to image generation falls in the category of latent variable models, represented as $p_\theta(\mathbf{x}_0) := \int p_\theta(\mathbf{x}_{0:T})\, d\mathbf{x}_{1:T}$ where the joint distribution $p_\theta(\mathbf{x}_{0:T})$ is referred to as the reverse process (or denoising process). This reverse process, modeled as a Markov chain, seeks to learn the original data distribution by inverting the diffusion process used during training, which gradually adds noise to the sample $\mathbf{x}_0$ according to the posterior $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$. Compared to GANs, diffusion models have shown more stability and controllability during training, as well as improved image quality [22].

The popular denoising diffusion probabilistic model (DDPM) proposed

by Ho et al. [40] explains the diffusion process (also called the forward flow) as a procedure that gradually adds Gaussian noise to corrupt the data $\mathbf{x}_0$ over time steps $t \in [0, T]$ eventually producing a fully noisy sample $\mathbf{x}_T$. This process uses the variance schedule $\beta_1, ..., \beta_T$.

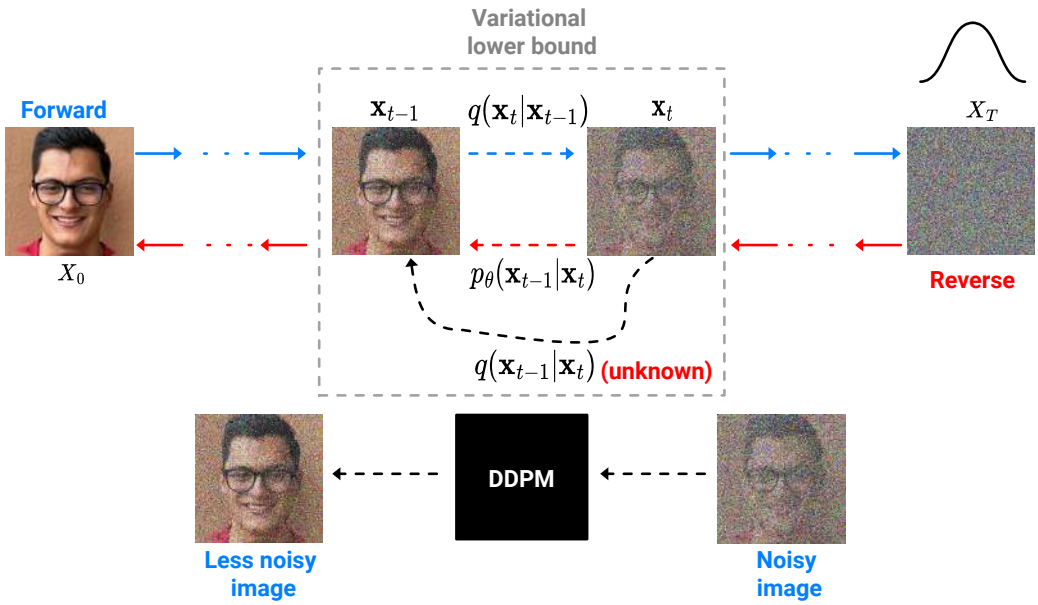


Figure 3.1: DDPM scheme. The forward diffusion process gradually corrupts a clean image x_0 with Gaussian noise over T timesteps until it becomes nearly pure noise x_T. The reverse process—parameterized by a neural network—is then trained to sequentially denoise x_T back toward x_0 by maximizing a variational lower bound on the data likelihood.

In DDPM framework, the forward process variances β_t are fixed to constants so, this posterior q has no learnable parameters, as shown in Figure 3.1, and formally defined by:

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) := \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}), \tag{3.1}$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\,\mathbf{x}_{t-1}, \beta_t \mathbf{I}). \tag{3.2}$$

A very important property of the forward process is that it allows sampling of data x_t at any time step t using the closed-form expression:

$$q(\mathbf{x}_T|\mathbf{x}_0) := \mathcal{N}\left(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right) \tag{3.3}$$

where $\alpha_t := 1 - \beta_t$ and $\quad \bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s$.

In contrast, the reverse process starts with an image full of noise x_T from $p(x_T) = \mathcal{N}(x_T; \mathbf{0}, \mathbf{I})$ and attempts to reconstruct the original data as follows:

$$p_\theta(\mathbf{x}_{0:T}) := p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \tag{3.4}$$

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2\mathbf{I}) \tag{3.5}$$

Given that β_t values are constants, σ_t^2 can be set to $\sigma_t^2 = \beta_t$ for initialization at a fully noisy state $\mathbf{x}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Additionally, $\sigma_t^2 = \tilde{\beta}_t$, where $\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$ helps to ensure that $\mathbf{x}_t$ is deterministically mapped to a single specific point.

To effectively estimate $\mu_\theta(\mathbf{x}_t, t)$, Ho et al. [40] proposed training an estimator $\epsilon_\theta(\cdot)$ using the following variational bound:

$$L(\theta) := \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\right] \tag{3.6}$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. This estimator $\epsilon_\theta(\cdot)$ helps to predict the noise level in the image at each time step, allowing the model to produce progressively less noisy images. This denoising process is also known as sampling process, where $\mathbf{x}_{t-1} \sim p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$, computed as:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(\mathbf{x}_t, t)\right) + \sigma_t\epsilon \tag{3.7}$$

3.2 Denoising Diffusion Implicit Model

A faster generation scheme was proposed in the denoising diffusion implicit model (DDIM) by Song et al [53]. In this scheme they diverge from the Markovian guideline of DDPM more specifically in the equation (3.7). This approach extends the inference process with the main objective to reduce

the number of steps of the reverse process once the model is trained, in other words, they train the model using the same logic as DDPM [53] but in inference time they use fewer steps for sampling x_0. The forward process is defined as follows:

$$q_\sigma(x_{1:T} \mid x_0) = q_\sigma(x_T \mid x_0) \prod_{t=2}^{T} q_\sigma(x_{t-1} \mid x_t, x_0), \tag{3.8}$$

$$q_\sigma(x_T \mid x_0) = \mathcal{N}\big(\sqrt{\alpha_T}\, x_0,\ (1-\alpha_T)I\big) \ \textit{for all } t > 1, \tag{3.9}$$

$$q_\sigma(x_{t-1} \mid x_t, x_0) = \mathcal{N}\Big(\sqrt{\alpha_{t-1}}\, x_0 + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\ \frac{x_t - \sqrt{\alpha_t}\, x_0}{\sqrt{1-\alpha_t}},\ \sigma_t^2 I\Big) \tag{3.10}$$

$$q_\sigma\big(x_t \mid x_{t-1},\, x_0\big) = \frac{q_\sigma\big(x_{t-1} \mid x_t,\, x_0\big)\, q_\sigma\big(x_t \mid x_0\big)}{q_\sigma\big(x_{t-1} \mid x_0\big)} \ \textit{(Forward process from Bayes)} \tag{3.11}$$

We have colored x_0 and x_{t-1} to show how DDIM differs from DDPM sampling. In addition, here we have a family $\mathcal{Q}$ of inference distributions, indexed by a real vector $\sigma \in \mathbb{R}^T_{\geq 0}$. The parameter σ determines the degree of randomness in the forward diffusion process. In the limiting case $\sigma \to 0$, the process becomes completely deterministic: knowing x_0 and x_t for any t uniquely fixes the previous state x_{t-1}. Given this, the forward process is no longer Markovian since each x_t could depend on both x_{t-1} and x_0.

In the other hand, we have the reverse process modeled by the conditional distribution $q_\sigma(x_{t-1} \mid x_t, x_0)$ where the noisy observation x_t was used to get a prediction of x_0, then used to obtain the sample x_{t-1}. The prediction of x_0 given x_t can be computed as:

$$f_\theta^{(t)}(x_t) \ := \ \frac{x_t \ - \ \sqrt{1-\alpha_t}\ \epsilon_\theta^{(t)}(x_t)}{\sqrt{\alpha_t}}\,. \tag{3.12}$$

with this, the generative process with a fixed prior $p_\theta(x_T) = \mathcal{N}(0, I)$ is defined:

$$p_\theta^{(t)}(x_{t-1} \mid x_t) = \begin{cases} \mathcal{N}\big(f_\theta^{(1)}(x_1),\ \sigma_1^2 I\big), & t = 1, \\ q_\sigma\big(x_{t-1} \mid x_t,\ f_\theta^{(t)}(x_t)\big), & \text{otherwise.} \end{cases} \tag{3.13}$$

where $q_\sigma(x_{t-1} \mid x_t, f_\theta^{(t)}(x_t))$ is defined as in 3.10 having x_0 replaced by $f_\theta^{(t)}(x_t)$).

After being defined the forward and backward process in DDIM, the approach in training uses the same training objective and procedure used in DDPM [40]. Given this, the new sampling formula for DDIM looks as follows:

$$x_{t-1} = \underbrace{\sqrt{\alpha_{t-1}}\Big(\frac{x_t - \sqrt{1-\alpha_t}\,\epsilon_\theta^{(t)}(x_t)}{\sqrt{\alpha_t}}\Big)}_{\text{predicted } x_0} + \underbrace{\sqrt{1-\alpha_{t-1}-\sigma_t^2}\;\epsilon_\theta^{(t)}(x_t)}_{\text{direction pointing to } x_t} + \underbrace{\sigma_t\,\epsilon_t}_{\text{random noise}} \tag{3.14}$$

Having this new formulation, DDIM considers the forward process not on all latent variables $x_{1:T}$, but on a sequential subset $x_{\tau_1}, .., x_{\tau_S}$ of length S matching the marginals from 3.9 on $q(x_{\tau_i} \mid x_0)$. When the length of the S is smaller than T, it accelerates the generative process, reducing the number of steps for inference without having to train a different model as seen in Figure 3.2.

3.3 Diffusion Models with Iteratively Preconditioned Guidance

Denoising diffusion models (DDMs) have recently emerged as powerful, zero-shot priors for solving inverse imaging problems—such as deblurring or super-resolution—by iteratively injecting and removing Gaussian noise. In parallel, plug-and-play (PnP) methods replace proximal mappings with pre-trained Gaussian denoisers, enforcing measurement consistency via least-squares (LS) or back-projection (BP) steps [54]. Likewise, guided diffusion sampling steers the reverse diffusion process toward agreement with observations by adding data-fidelity gradients to each denoising update [55]. Building on these strands, Garber and Tirer introduce Denoising Diffusion with iteratively Preconditioned Guidance (DDPG) [25]: a unified sampling framework that, at each diffusion step, applies an iteration-dependent preconditioner to smoothly transition from BP-based pseudoinverse guidance in

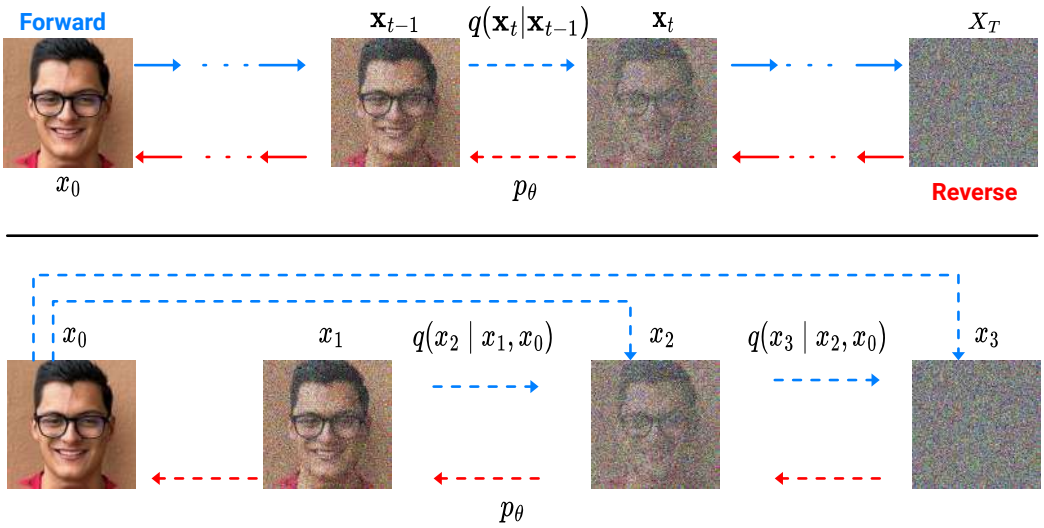


Figure 3.2: Comparison of DDPM (top) and DDIM (bottom). The DDPM includes a series of consecutive steps in both the forward and reverse process, forming a fully Markovian chain of noise addition and removal. By contrast, DDIM shares the same forward noise schedule but replaces the reverse chain with a non-Markovian, deterministic update that can skip intermediate timesteps for much faster sampling without retraining.

early iterations to LS guidance in later ones. By combining the fast convergence and strong data consistency of BP with the noise robustness of LS, DDPG achieves both high accuracy and superior perceptual quality in restored images.

In image restoration, the observed measurement $\mathbf{y}$ is modeled as

$$\mathbf{y} = A\,\mathbf{x}^* + \mathbf{e}, \tag{3.15}$$

where $\mathbf{x}^*$ denotes the unknown clean image, A is a known degradation operator (e.g., blur, downsampling, or masking), and $\mathbf{e}$ represents additive measurement noise. The goal of any restoration method is to recover an accurate estimate of $\mathbf{x}^*$ from the corrupted observation $\mathbf{y}$ by "inverting" the effect of A while mitigating the influence of $\mathbf{e}$, typically through prior models or

learned denoisers that regularize this ill-posed inverse problem.

This approach combines the DDIM logic with data-fidelity guidance vectors to achieve perceptually pleasing restorations. Specifically, they define the following guidance vectors:

$$\mathbf{g}_{BP} = \mathbf{A}^T \left(\mathbf{A}\mathbf{A}^T + \eta \mathbf{I}_m\right)^{-1} \left(\mathbf{A}\mathbf{x}_{0|t} - \mathbf{y}\right)$$
$$\mathbf{g}_{LS} = c\mathbf{A}^T \left(\mathbf{A}\mathbf{x}_{0|t} - \mathbf{y}\right)$$

where $c = 1$ and $\eta > 0$ is a regularization parameter [54, 56]. These two guidance vectors ($\mathbf{g}_{BP}$, $\mathbf{g}_{LS}$) correspond to a Back Projection (BP) and Least Squares (LS) gradients, respectively, applied to the intermediate estimates during the inverse process. The BP vector also known as pseudoinverse or range/null-space guidance [54], has demonstrated superior consistency with $\mathbf{y}$ (de-gradated version of the target image to be reconstructed) and requires fewer iterations to denoise images in Plug-and-Play (PnP) [57]. These methods combine combine problem-agnostic diffusion models with the measurement process to solve inverse problems [55]. On the other hand, LS is better than BP in the noisier stages [58, 59, 25]. The DDPG approach also proposes an engineered solution to apply these data fidelity guidance vectors in the following way:

$$\tilde{\mathbf{x}}_{t-1} = \mathbf{x}_{0|t} - \mu_t \left((1-\delta_t)\mathbf{g}_{BP}(\mathbf{x}_{0|t}) + \delta_t \mathbf{g}_{LS}(\mathbf{x}_{0|t})\right) \tag{3.16}$$

where μ_t is set to 1 and $0 \leq \delta_T < \cdots < \delta_0 \leq 1$ helps to transition between $\mathbf{g}_{BP}$ (used more in early iterations) and $\mathbf{g}_{LS}$ (prioritized in later iterations). Additionally, in the sampling scheme, DDPG proposes several modifications to the additive noise terms $\hat{\sigma}_t \epsilon_\theta(\mathbf{x}_t; t) + \tilde{\sigma}_t \epsilon_t$ from the DDIM approach, as described in equation 3.14. They suggest that the guidance can significantly alter the signal, causing the effective predicted noise to deviate from the term $\epsilon_\theta(\mathbf{x}_t; t)$ in DDIM. Therefore, they propose the effective predicted noise formulation in terms of $\tilde{\mathbf{x}}_{t-1}$ as follows:

$$\hat{\epsilon}_t = \frac{1}{\sqrt{1-\bar{\alpha}_t}} \left(x_t - \sqrt{\bar{\alpha}_t}\tilde{x}_{t-1}\right) \tag{3.17}$$

Another important modification made in the DDPG approach is that when $\mathbf{g}_{BP}$ is predominant in the initial iterations, the effective predicted noise $\hat{\epsilon}_t$ may not be useful due to its sensitivity to the noise contained in $\mathbf{g}_{BP}$. These results led to a simplified method to transition between the effective estimated noise ($\hat{\epsilon}_t$) and the stochastic noise $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$. Instead of using $\hat{\sigma}_t$, they use a simple tuning weight of $\sqrt{1-\zeta}$ and $\sqrt{\zeta}$, where $\zeta \in [0, 1]$. Following these updates of the equation 3.14, DDPG presents the updated sampling equation as follows:

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\tilde{\mathbf{x}}_{t-1} + \sqrt{1-\bar{\alpha}_{t-1}}\left(w_t\sqrt{1-\hat{\zeta}}\hat{\epsilon}_t + \sqrt{\hat{\zeta}}\epsilon_t\right) \tag{3.18}$$

For a complete overview of how these updates integrate into the original DDPG algorithm, refer to the following section:

Algorithm 1 DDPG

Input: Noise estimator $\epsilon_\theta(\cdot, t)$, T, y, A, $\{\bar{\alpha}_t\}$, $\{\mu_t\}$, c, η, γ, ζ, δ_t
Initialize $x_T \sim \mathcal{N}(0, I_n)$
for $t = T$ **to** 1 **do**
  $x_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(x_t, t))$
  $g_{BP} = A^T(AA^T + \eta I_m)^{-1}(Ax_{0|t} - y)$
  $g_{LS} = cA^T(Ax_{0|t} - y)$
  $\tilde{x}_{t-1} = x_{0|t} - \mu_t\big((1-\delta_t)g_{BP} + \delta_t g_{LS}\big)$
  $\hat{\epsilon}_t = \frac{1}{\sqrt{1-\bar{\alpha}_t}}(x_t - \sqrt{\bar{\alpha}_t}\tilde{x}_{t-1})$
  Sample $\hat{\epsilon} \sim \mathcal{N}(0, I_n)$
  $x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\tilde{x}_{t-1} + \sqrt{1-\bar{\alpha}_{t-1}}\big(w_t\sqrt{1-\zeta}\hat{\epsilon}_t + \sqrt{\zeta}\epsilon_t\big)$
end for

An opportunity arises from manipulating the data fidelity vectors ($\mathbf{g}_{BP}$ and $\mathbf{g}_{LS}$) when guiding the reverse sampling process. In regards to face recognition, these vectors show how closely our facial image approximates the target image. We empirically manipulate the identity features of these vectors to obtain a controllable de-identified version of the target subject based on different denoising trajectories compared to the vanilla image restoration.

Chapter 4

Masking identities: Customizable Face Anonymization with Diffusion Models

In this chapter, we state MaskDDPG, a novel framework that builds on the DDPG sampler to deliver:

- Controllable privacy via a user-defined parameter $\lambda \in [0, 1]$that modulates guidance strength.
- Smooth trade-off between image quality and anonymization.
- Spatial selectivity through a segmentation mask that designates exactly which facial regions to deidentify, preserving soft-biometric attributes (e.g., background, glasses, expression).

4.1 Overview of MaskDDPG

In this section, we present the novel MaskDDPG framework, which builds on the generative capabilities of the modern diffusion model DDPG and offers

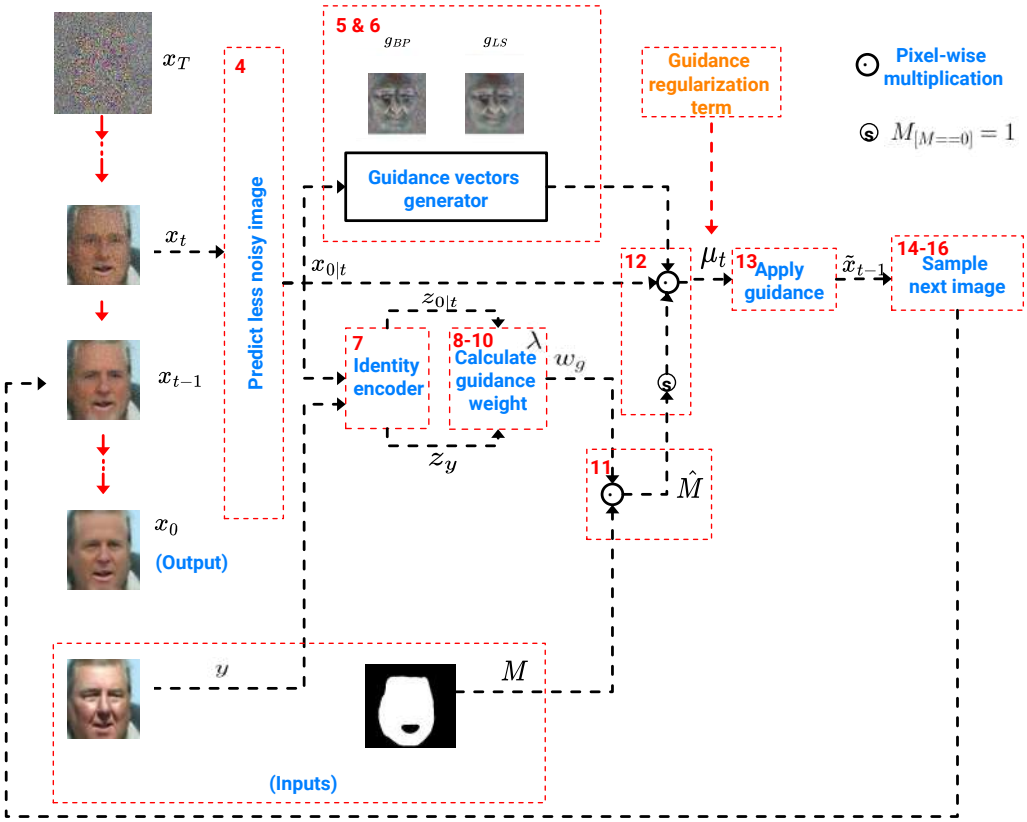


Figure 4.1: MaskDDPG Pipeline. For reference, at the top left of the highlighted boxes can be found the corresponding line number from the Algorithm 2.

multiple benefits over its competitors, including performance competitive to the state-of-the-art for face deidentification, controllable privacy levels, and high-quality anonymized images. A high-level overview of our model is presented in Figure 4.1; for implementation details, refer to Algorithm 2.

MaskDDPG takes a face image y and iteratively refines a noisy estimate x_t to produce a partially anonymized image. At each timestep t the model follows these steps for image generation:

1. The model estimates the clean image $x_{0|t}$ from the current noised input x_t using the denoiser $\epsilon_\theta(\cdot, t)$.

2. It computes two guidance terms, g_{BP} and g_{LS}, from $x_{0|t}$ and measurement y.

3. Simultaneously, identity embeddings $z_{0|t} = \mathrm{ID}(x_{0|t})$ and $z_y = \mathrm{ID}(y)$ are extracted via a pretrained TransFace encoder [60].

4. A user-defined privacy parameter $\lambda \in (0, 1]$ modulates the similarity-based guidance weight w_g, balancing identity divergence and feature preservation.

5. We generate a mask M indicating regions for deidentification, multiply it by w_g to obtain $\hat{M}$, and compute a guidance regularization map $\mu_t = \mathbb{I}(\hat{M} = 0) + \hat{M}$.

6. Finally, the denoising step applies the masked guidance: $\tilde{x}_{t-1} = x_{0|t} - \mu_t\big((1 - \delta_t)g_{BP} + \delta_t g_{LS}\big)$, and samples x_{t-1} accordingly.

For a complete overview of our approach, refer to the algorithm 2 where it shows the step by step how the previously described methodology works in a compact way.

4.1.1 Diffusion Model

MaskDDPG builds on the DDPG variant of the DDIM sampler [53], which introduces preconditioned guidance in the denoising update. Our key contribution is replacing the constant regularization in DDPG with a dynamic, mask-weighted guidance term μ_t, computed at each step. This mechanism encourages incremental identity shifts (weaker guidance) when embeddings are similar (small w_g) and applies less identity shifts (stronger guidance) when identities differ (large w_g).

The MaskDDPG diffusion takes a step further in a dynamic and automatic way, by promoting the sampling of new images to be distant in identity from the input while preserving non-identity features (soft biometrics) and

Algorithm 2 Masked Denoising Diffusion with Iterative Preconditioned Guidance (MaskDDPG)

Input: Noise estimator $\epsilon_\theta(\cdot, t)$, T, y, A, $\{\bar{\alpha}_t\}$, c, η, γ, ζ, δ_t, M, λ, z_y
Initialize $x_T \sim \mathcal{N}(0, I_n)$
for $t = T$ **to** 1 **do**
  $x_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(x_t, t))$
  $g_{BP} = A^T(AA^T + \eta I_m)^{-1}(Ax_{0|t} - y)$
  $g_{LS} = cA^T(Ax_{0|t} - y)$
  $z_{0|t} = identity_embedding(x_{0|t})$
  $sim = \text{cosine}(z_{0|t}, z_y)$,
  $\text{cutoff} = \frac{1}{1+\lambda}$,

$$w_g = \begin{cases} 1 - sim \cdot (1+\lambda), & sim \leq \text{cutoff} \\ 10^{-8}, & sim > \text{cutoff} \\ 1, & \text{if } \lambda = 0 \end{cases}$$

  $\hat{M} = M \cdot w_g$
  $\mu_t = \mathbb{I}(\hat{M} = 0) + \hat{M}$
  $\tilde{x}_{t-1} = x_{0|t} - \mu_t((1-\delta_t)g_{BP} + \delta_t(g_{LS}))$
  $\hat{\epsilon}_t = \frac{1}{\sqrt{1-\bar{\alpha}_t}}(x_t - \sqrt{\bar{\alpha}_t}\tilde{x}_{t-1})$
  Sample $\epsilon_t \sim \mathcal{N}(0, I_n)$
  $x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\tilde{x}_{t-1} + \sqrt{1-\bar{\alpha}_{t-1}}\big(w_t\sqrt{1-\zeta}\hat{\epsilon}_t + \sqrt{\zeta}\epsilon_t\big)$
end for

producing high-realistic images. By leveraging the pretrained DDPG denoiser and augmenting it with mask-based, identity-aware guidance, MaskDDPG synthesizes images that lie outside the original identity distribution without retraining or fine-tuning.

4.2 Identity Conditioning and Privacy Control

The model starts by generating Gaussian Noise in the same shape as the images that the model used to be pretrained. This Gaussian noise is fed into the U-Net noise predictor, $\epsilon_\theta(\cdot, t)$, to predict the noise component in the input. This noise prediction is used to produce a less-noisy image, $x_{0|t}$ as shown in figure 4.2. Once the less-noisy sample $x_{0|t}$ at time step t is obtained, we extract its identity features for conditioning. We then compute the guidance vectors (g_{BP}, g_{LS} from DDPG implementation) using the observation y, the image to be deidentified.

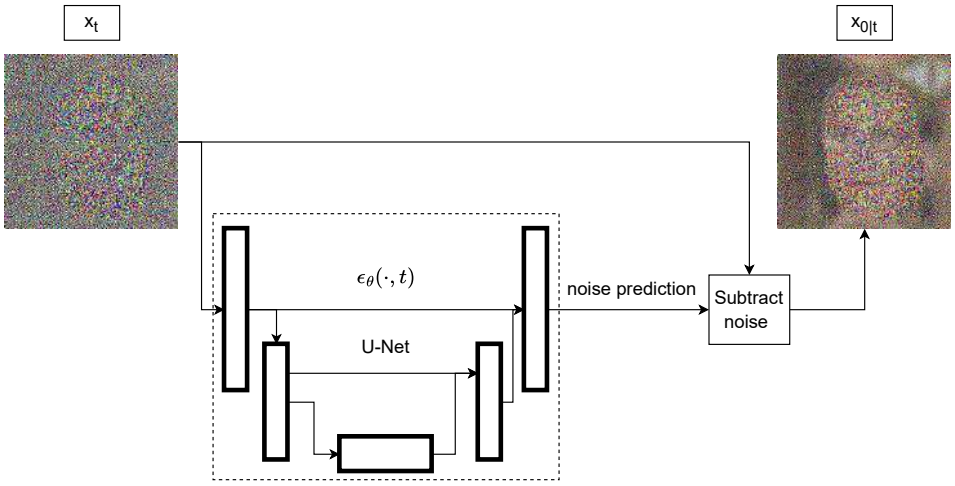


Figure 4.2: Less noisy image prediction example ($x_{0|t}$).

The main idea of identity conditioning in this approach is to increase the distance between the identity of the input image (y) and the newly sampled image ($x_{0|t}$) at each diffusion reverse step t. This similarity between samples is computed by using the identity features of both images obtained from the AdaFace [28] encoder. The identity features (z_y) from the original image y are computed before the diffusion process and are part of the input parameter of this approach. In each diffusion step, the identity features ($z_{0|t}$) from the

less noisy image ($x_{0|t}$) are produced to compute the cosine similarity between both:

$$sim = \frac{\sum_{i=1}^{n}(z_{0|t})_i(z_y)_i}{\sqrt{\sum_{i=1}^{n}(z_{0|t})_i^2} \cdot \sqrt{\sum_{i=1}^{n}(z_y)_i^2}} \tag{4.1}$$

There is an important cutoff here that needs to be computed, which is $\frac{1}{1+\lambda}$, where serves as an upper bound for the similarity allowed between samples at each time step t given a user parameter $\lambda \in [0, 1]$. Once this similarity measure is computed (standardized to have values between $[0, 1]$) as well as the cutoff, the algorithm can modify the sampling process of the diffusion model at any specific t to go always against the direction of the similarity measure. This is done by regularizing the weight w_g which tells the algorithm how much guidance should be removed to preserve both images to be distanced. This weight is a dynamically changing during the diffusion process: it decreases the guidance when the identity similarity between y and $x_{0|t}$ is high, and relaxes it when the similarity is low. This approach is applied at different levels $\lambda \in [0, 1]$.

To compute the guidance weight w_g, we use:

$$w_g = \begin{cases} 1 - sim \cdot (1 + \lambda), & sim \leq \text{cutoff} \\ 10^{-8}, & sim > \text{cutoff} \\ 1, & \text{if } \lambda = 0 \end{cases} \tag{4.2}$$

If $\lambda = 0$, we set $w_g = 1$, which means the guidance stays unchanged, outputting a restored version of the original image (vanilla DDPG) and preserving the original identity. This weight w_g can take different values bellow 0.5, constrained to the input parameter λ as shown in the following figure 4.3.

Figure 4.3 shows how, for each λ, the available weight space is constrained across similarity values. Beyond the cutoff, weights drop to near zero (10^{-8}), effectively canceling guidance.

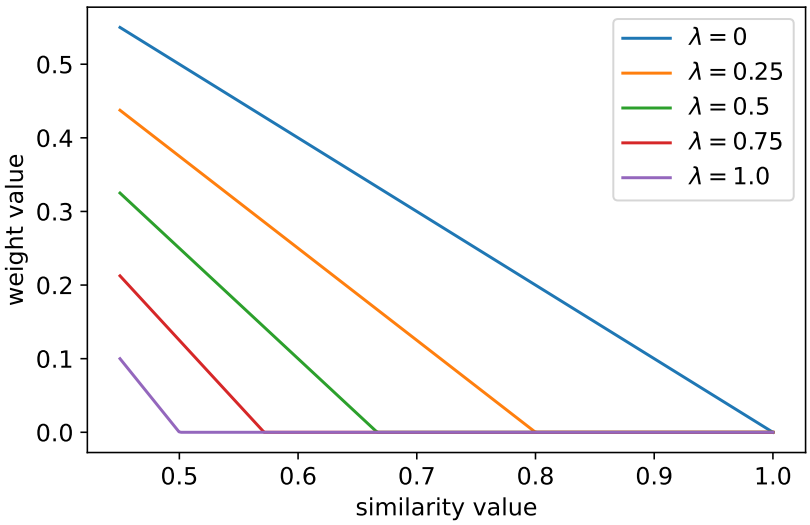


Figure 4.3: Influence of the weight w_g at different similarity values and constrained to different λ values.

4.3 Mask conditioning

Once this weight is computed, it is going to be multiplied against a binary mask (M) to make a spacial regularization as shown in figure 4.4. This mask M depicts parts of the face from the input image y which marks with 1's where this value w_g needs to be placed and 0's otherwise.

Figure 4.4: Point-wise multiplication of the guidance term w_g to the mask (M).

For example, if the user wants to protect the nose, eyes and skin from the

face these regions are marked with 1 in the mask and after a multiplication against our weight ($M \cdot w_g$), this regions will take on the value w_g. The facial parts that were not intended to be protected (no deidentification), the value will keep being 0. After multiplication, we denote the new version of the mask as $\hat{M}$.

Once we have $\hat{M}$, we have to convert all the values that are 0's to 1's. The regions of the mask that originally had 0 as value, are the regions we do not want to update in the image, meaning that the full magnitude and direction of the guidance will be kept to avoid reducing guidance from parts where the deidentification is not needed. Once we have done all this we end up with the final regularization term μ_t defined as:

$$\mu_t = \mathbb{I}(\hat{M} = 0) + \hat{M}. \tag{4.3}$$

This spatial map μ_t modulates the combined guidance $(1-\delta_t)g_{BP}+\delta_t g_{LS}$, reducing it in masked regions while preserving it elsewhere as shown in the figure 4.5.

4.4 Guidance Application and New Image Sampling

When the regularization is applied to the guidance tensor $\mu_t\big((1-\delta_t)\, g_{BP} + \delta_t\, g_{LS}\big)$, the model subtracts this combined term from the sample $x_{0|t}$. By doing so, the model not only produces a less noisy image, but also guides its generation according to the specified guidance (see Figure 4.6). A moving average between g_{BP} and g_{LS} determines which of the two terms dominates the guidance during denoising. At the beginning of the process ($t \geq 70$), g_{LS} is in control; thereafter, until step 0, g_{BP} takes over. Garber et al. [25] demonstrate that g_{BP} is more effective in the later, less noisy denoising steps.

Another important property of this mechanism is that, at each time step t, it enables traversal through different directions in the data distribution,

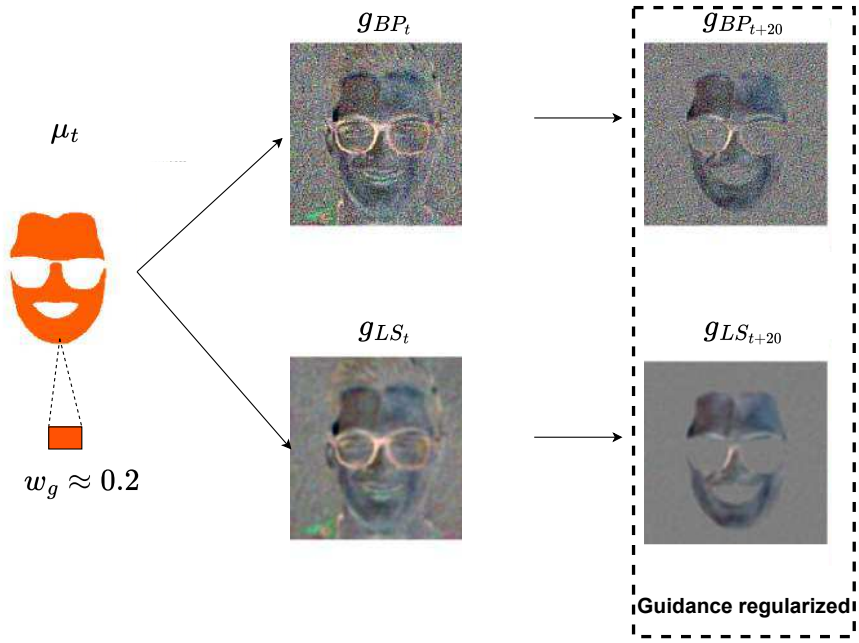


Figure 4.5: Point-wise multiplication of the guidance term μ_t to the guidance terms g_{BP}, g_{LS} when the identity from the original subject y needs to be removed by spatially regularizing the guidance terms through generation time.

yielding identities that are progressively closer to or further from the original, as illustrated in Figure 4.7. When λ exceeds one, we observe degraded performance, since the estimated images fall out-of-distribution due to the negative values introduced in the weight w_g.

The final step is to sample a new image for the next time step—or, if $t = 0$, to produce the final output—by reintroducing a small fraction of noise via the Gaussian trick. First, we compute the noise extracted from x_t:

$$\hat{\epsilon}_t = \frac{x_t - \sqrt{\overline{\alpha}_t}\,\tilde{x}_{t-1}}{\sqrt{1 - \overline{\alpha}_t}}. \tag{4.4}$$

Next, we sample fresh Gaussian noise $\epsilon_t \sim \mathcal{N}(0, I_n)$ and add it to $\tilde{x}_{t-1}$ to generate the new image x_{t-1}. This procedure is then repeated until the denoising process is complete.

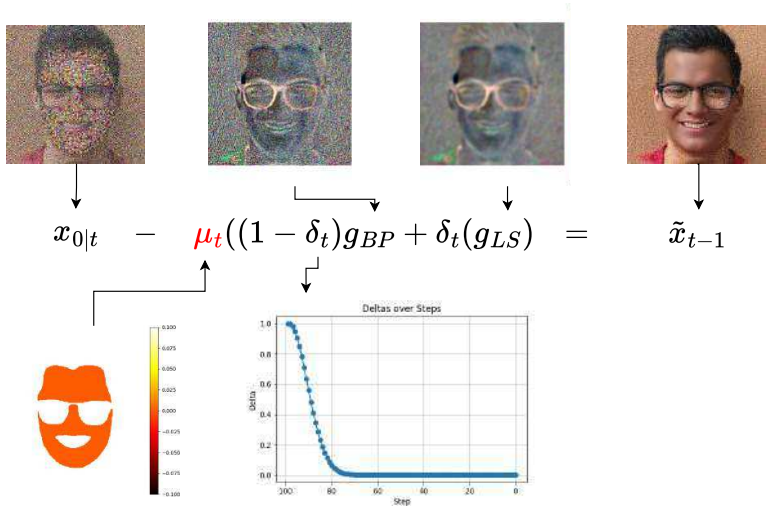


Figure 4.6: Overview of the MaskDDPG guidance process for de-identification. A moving average balances the BP and LS guidance vectors applied to our mask μ_t, weighted by δ_t, and is subtracted from the current estimate $x_{0|t}$ to produce the refined output $\tilde{x}_{t-1}$.

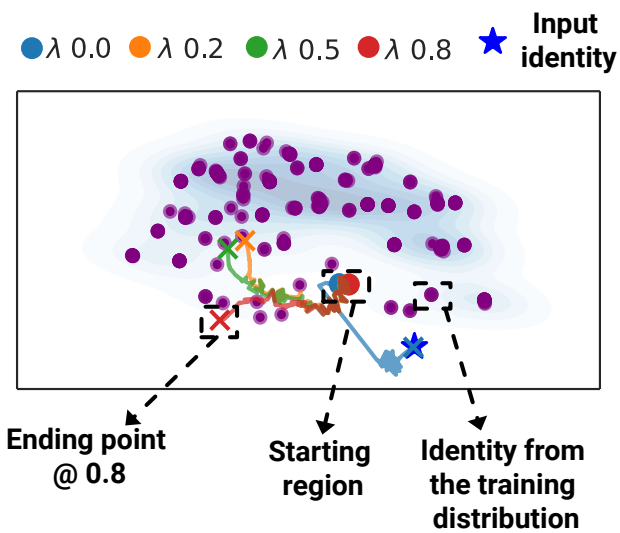


Figure 4.7: Markov chains for different λ values used in our approach *w.r.t* the identity space (obtained from T-SNE) from the training distribution.

Chapter 5

GauDeiDiff: Gaussian kernels for face Deidentification in Diffusion Models

5.1 Overview of GauDeiDiff

In this section, we present the GauDeiDiff framework, which builds on the modern diffusion approach with preconditioned guidance called DDPG [25]. This framework offers multiple benefits, including state-of-the-art privacy protection techniques and high-quality image generation. A high-level overview of the method is shown in Figure 5.1, and a detailed description of the process is given in Algorithm 3.

GauDeiDiff is a guided diffusion sampler designed to produce de-identified facial images by integrating per-step spatial kernels into the standard preconditioned guidance framework. At a high level, the algorithm performs the following stages:

1. **Initialization.** We start with a small batch of two independent Gaussian noise samples

$$\{x_T^{(1)}, x_T^{(2)}\} \sim \mathcal{N}(0, I_n),$$

Algorithm 3 GauDeiDiff algorithm

Input: Noise estimator $\epsilon_\theta(\cdot, t)$, T, y, A, $\{\bar{\alpha}_t\}$, $\{\mu_t\}$, c, η, γ, ζ, δ_t, bbox, λ, t_μ, $vars$
Initialize batch $x_t = [\, x_T^{(1)},\, x_T^{(2)} \,]$, where each $x_T^{(i)} \sim \mathcal{N}(0, I_n)$
for $t = T$ **to** 1 **do**
  $x_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(x_t, t))$
  $GK = \text{get_GK}(\text{shape}(x_{0|t}), vars[t], bbox)$
  $LoG = \text{get_LoG}(\text{shape}(x_{0|t}), vars[t], bbox)$
  if $t = t_\mu$ **then**
    $x_{0|t_\mu} = \text{mean}\big(\{x_{0|t}^{(i)}\}_{i=1}^{2}\big)$
    $\{x_{0|t}^{(i)}\}_{i=1}^{3} \;=\; \{x_{0|t}^{(1)},\, x_{0|t}^{(2)},\, x_{0|t_\mu}\}$
    $zero_K = \text{get_zero_k}\big(\text{shape}(x_{0|t}), \text{bbox}\big)$
    $\{K^{(i)}\}_{i=1}^{3} \;=\; \{GK^{(1)},\, LoG^{(2)},\, zero_K\}$
  else
    $\{K^{(i)}\}_{i=1}^{2} \;=\; \{GK^{(1)},\, LoG^{(2)}\}$
  end if
  $g_{BP} = A^T(AA^T + \eta I_m)^{-1}(Ax_{0|t} - y)$
  $g_{LS} = cA^T(Ax_{0|t} - y)$
  Apply per–kernel masking: $g_{BP}^{(i)} \leftarrow g_{BP}^{(i)} \odot K^{(i)}, \quad g_{LS}^{(i)} \leftarrow g_{LS}^{(i)} \odot K^{(i)}$
  $\tilde{x}_{t-1} = x_{0|t} - \mu_t\big((1-\delta_t)g_{BP} + \delta_t g_{LS}\big)$
  if $t = t_\mu$ **then**
    $\hat{\epsilon}_t = \{\frac{1}{\sqrt{1-\bar{\alpha}_t}}(x_t^{(i)} - \sqrt{\bar{\alpha}_t}\tilde{x}_{t-1}^{(i)})\}_{i=1}^{2}$
    $\hat{\epsilon}_t^{\mu} = \frac{1}{2}\big(\hat{\epsilon}_t^{(1)} + \hat{\epsilon}_t^{(2)}\big)$
    $\hat{\epsilon}_t \;\leftarrow\; \{\hat{\epsilon}_t^{(1)},\, \hat{\epsilon}_t^{(2)},\, \hat{\epsilon}_t^{\mu}\}$
  else
    $\hat{\epsilon}_t = \frac{1}{\sqrt{1-\bar{\alpha}_t}}(x_t - \sqrt{\bar{\alpha}_t}\tilde{x}_{t-1})$
  end if
  Sample $\{\epsilon^{(i)}\}_{i=1}^{s} \sim \mathcal{N}(0, I_n)$
  $x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\tilde{x}_{t-1} + \sqrt{1-\bar{\alpha}_{t-1}}\big(w_t\sqrt{1-\zeta}\hat{\epsilon}_t + \sqrt{\zeta}\epsilon\big)$
end for

and determine a special time index t_μ at which to inject an "average-face" inference via a percentile parameter λ. We also precompute a schedule of kernel variances over the diffusion process.

2. **Reverse Diffusion Loop.** For each timestep $t = T, T-1, \ldots, 1$:

   - *Denoised Estimate.* Compute the estimate of the clean image for each batch member,

$$x_{0|t}^{(i)} = \frac{x_t^{(i)} - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(x_t^{(i)}, t)}{\sqrt{\bar{\alpha}_t}}, \quad i = 1, 2.$$

   - *Kernel Construction.* Generate two spatial weight maps restricted to the face bounding box:

$$\begin{aligned} GK^{(i)} &= \text{get_GK}\big(\text{shape}(x_{0|t}), \text{vars}[t], \text{bbox}\big), \\ LoG^{(i)} &= \text{get_LoG}\big(\text{shape}(x_{0|t}), \text{vars}[t], \text{bbox}\big). \end{aligned}$$

   - *Average-Face Injection.* At $t = t_\mu$, compute the mean over the two denoised estimates,

$$x_{0|t_\mu} \;=\; \text{mean}\big(\{x_{0|t}^{(1)},\, x_{0|t}^{(2)}\}\big),$$

     append it as a third sample, and introduce a third mask *zero_K* that nullifies updates inside the face region. The kernel set then expands from $\{GK,\, LoG\}$ to $\{GK,\, LoG,\, zero_K\}$.

   - *Guidance Computation.* Form the back-projection and least-squares guidance terms on the (possibly enlarged) batch:

$$g_{BP} = A^T(AA^T + \eta I)^{-1}\big(A\,X_{0|t} - y\big), \quad g_{LS} = c\,A^T\big(A\,X_{0|t} - y\big).$$

   - *Masking.* Apply each kernel $K^{(i)}$ element-wise to its corresponding guidance:

$$g_{BP}^{(i)} \leftarrow g_{BP}^{(i)} \odot K^{(i)}, \quad g_{LS}^{(i)} \leftarrow g_{LS}^{(i)} \odot K^{(i)}.$$

- *Preconditioned Update.* Combine the masked guidance to form the next denoised state,

$$\tilde{X}_{t-1} = X_{0|t} \; - \; \mu_t\Big((1-\delta_t)\, g_{BP} \; + \; \delta_t\, g_{LS}\Big).$$

- *Noise-Injection Step.* Compute the noise residual $\hat{\epsilon}_t$, sample fresh Gaussian noise ϵ_t, and perform the final update

$$X_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\, \tilde{X}_{t-1} + \sqrt{1-\bar{\alpha}_{t-1}}\Big(w_t \sqrt{1-\zeta}\, \hat{\epsilon}_t \; + \; \sqrt{\zeta}\, \epsilon_t\Big).$$

3. **Output.** After completing all T steps, the batch X_0 contains up to three de-identified reconstructions: two samples guided by Gaussian or Laplacian-of-Gaussian masks, and (if enabled) one "average-face" sample with zero-kernel masking.

This overview highlights how GauDeiDiff seamlessly incorporates spatial deidentification masks and an average-face branch into a standard preconditioned diffusion sampler, yielding controlled privacy protection without retraining the underlying deep learning denoiser as shown in the figure 5.1.

5.2 Incremental variance for the spatial kernels

The spatial kernels GK (Gaussian kernel) and LoG (Laplacian of Gaussian) lie at the heart of GauDeiDiff's deidentification strategy. Both kernels share the same variance parameter σ^2 (computed per-step according to the face bounding box), but their shapes differ: the Gaussian kernel is broader, yielding a higher level of deidentification, while the LoG kernel is narrower and preserves more detail.

Concretely, let (x_c, y_c) be the center of the face bounding box and x, y be the coordinates of the pixels inside it. We define

$$\mathrm{GK}(x,y) = 1 - \exp\Big(-\frac{(x-x_c)^2 + (y-y_c)^2}{2\,\sigma^2}\Big)$$

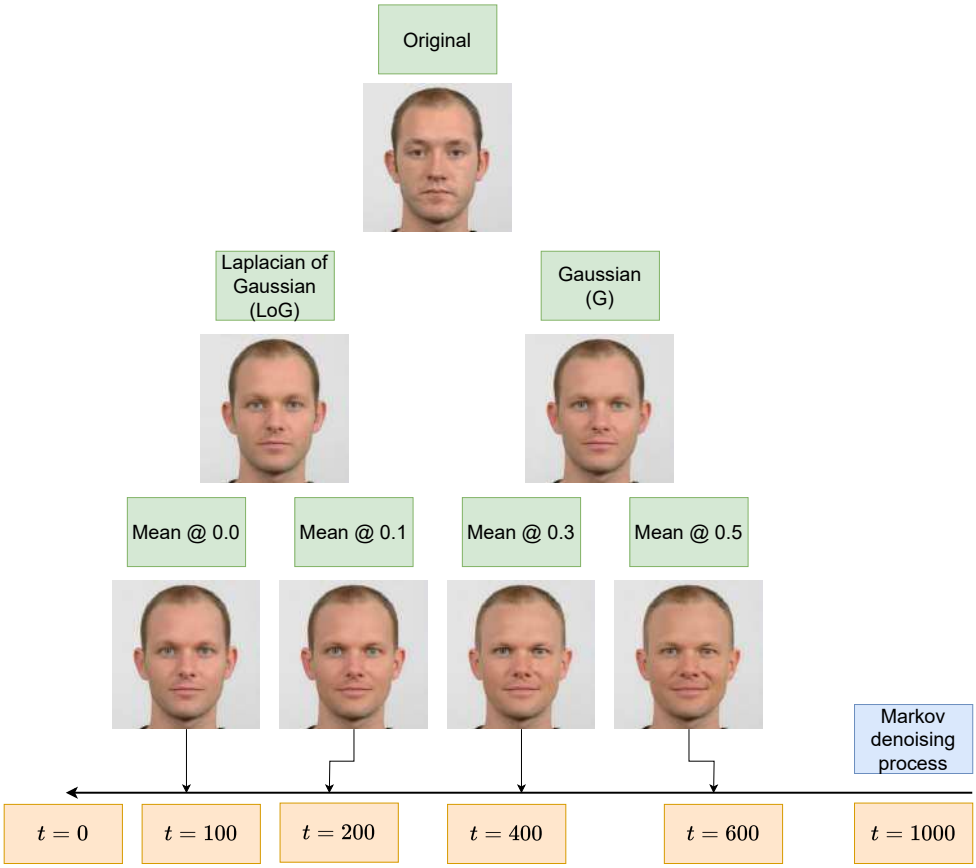


Figure 5.1: GauDeiDiff: Face deidentification using Gaussian kernels, Laplacian of Gaussian, and the mean inference from these kernels at any point in the Markov denoising process.

which inverts a centered Gaussian so that pixels at the box center are most strongly attenuated.

The Laplacian-of-Gaussian (Mexican-hat) kernel is

$$\mathrm{LoG}(x, y) = 1 - \left(1 - \frac{(x - x_c)^2 + (y - y_c)^2}{2\,\sigma^2}\right) \exp\left(-\frac{(x - x_c)^2 + (y - y_c)^2}{2\,\sigma^2}\right)$$

which produces a narrower attenuation profile: it first rises from the center before decaying, emphasizing a tighter face region.

In our implementation, the variance σ^2 is not fixed but increases linearly over the diffusion timesteps in accordance with the size of the detected face.

Let W and H be the width and height of the bounding box, and denote the total number of diffusion steps by T. We first set

$$\sigma_{\min}^2 = 0.01\,(W\,H), \qquad \sigma_{\max}^2 = W\,H,$$

where 0.01 ensures a small initial variance. Then for each step $t = 0, \ldots, T-1$ we compute

$$\sigma_k^2 = \sigma_{\min}^2 + \frac{\sigma_{\max}^2 - \sigma_{\min}^2}{T-1}\,t,$$

so that the kernel broadens incrementally from a tight, low-variance kernel to a broad, high-variance kernel as diffusion proceeds. Optionally, a single σ^2 value may be sampled from $\{\sigma_k^2\}$ when constructing a fixed-variance kernel.

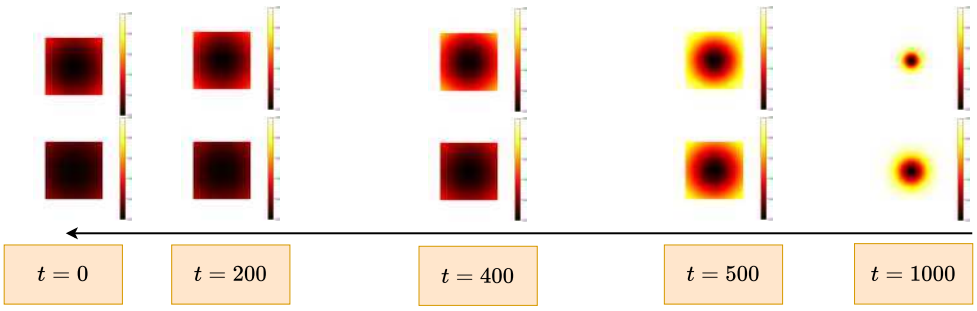


Figure 5.2: Incremental variance for GK (bottom) and LoG (top)

By gradually increasing σ^2 over the diffusion steps, we obtain an "incremental variance" schedule (as shown in figure 5.2) that yields two artificial samples at distinct privacy levels—one from the broad Gaussian mask and one from the sharper LoG mask—without altering the underlying diffusion model.

5.3 Application of the kernels over the guidance mechanism

Given the kernels GK and LoG, with their incremental variance schedule, we can sample two images in inference time (one per kernel), as shown in Figure 5.3:

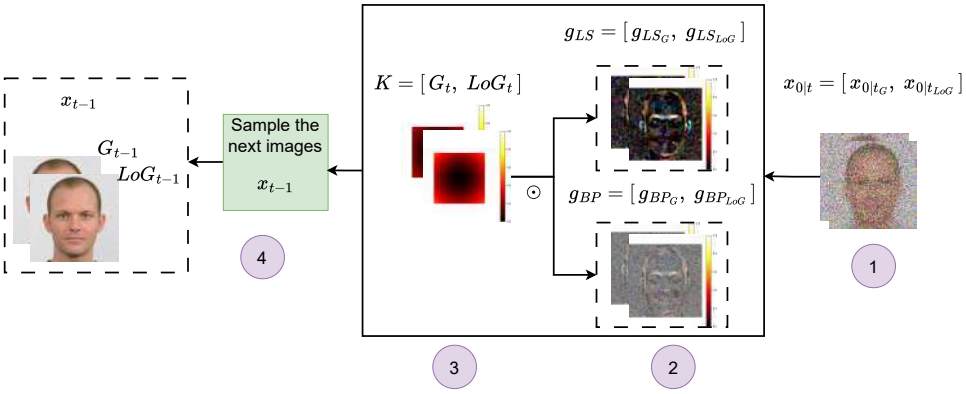


Figure 5.3: Image sampling when the "average-face" inference has not yet been produced.

To describe the figure, we illustrate the computation steps (purple circles with numbers) in this process. In the first step, we compute the denoised estimates at the current timestep. In the second step, we compute the guidance tensors for back-projection (BP) and least-squares (LS). Since the denoised estimates form a batch of two, we obtain two BP tensors and two LS tensors. In the third step, we apply the GK and LoG kernels by point-wise multiplying each guidance tensor by its corresponding mask. Once these kernels have been applied, we perform the preconditioned update:

$$\tilde{X}_{t-1} = X_{0|t} \ - \ \mu_t\big((1-\delta_t)\, g_{BP} + \delta_t\, g_{LS}\big).$$

This preconditioned update yields a batch of two intermediate tensors, each guided by its masked BP and LS. Finally, from the noise residual $\hat{\epsilon}_t$, we sample fresh Gaussian noise ϵ and perform the final update for the next iteration:

$$X_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\, \tilde{X}_{t-1} + \sqrt{1-\bar{\alpha}_{t-1}}\Big(w_t\sqrt{1-\zeta}\, \hat{\epsilon}_t + \sqrt{\zeta}\, \epsilon\Big).$$

With this batch sampling, we obtain two de-identified images whose degree of obfuscation differs according to the variance profiles of the Gaussian and Laplacian-of-Gaussian kernels. Specifically, the Gaussian-masked image exhibits stronger attenuation over the face region (i.e. larger central weights

and smoother fall-off), which more aggressively blurs and deidentifies facial features. In contrast, the LoG mask rises more sharply near the boundary before decaying, preserving more fine structure and thus resulting in a subtler level of obfuscation.

Concretely, at each denoising step we maintain a mini-batch of one sample guided by GK and one guided by LoG. For each sample $i \in \{GK,\, LoG\}$ we compute the back-projection and least-squares guidance tensors $g_{BP}^{(i)}$, $g_{LS}^{(i)}$, multiply element-wise by the corresponding mask $K^{(i)}$, and then apply the preconditioned update:

$$\tilde{x}_{t-1}^{(i)} = x_{0|t}^{(i)} \;-\; \mu_t\Big((1-\delta_t)\, g_{BP}^{(i)} + \delta_t\, g_{LS}^{(i)}\Big).$$

These two updated tensors $\{x_{t-1}^{(G)},\, x_{t-1}^{(LoG)}\}$ form the input batch for the next timestep $t-1$, ensuring that both deidentification levels evolve in parallel throughout the diffusion process.

5.4 User-specified privacy protection – average inference

With the parameter λ, the user specifies the desired privacy level of the "average-face" image. Concretely, given size of the set of all the time steps $T = \big|\{t_k\}\big|$ where $k \geq 2$ we compute a percentile index via the function

$$t_\mu \;= \texttt{get_percentile_index}(\{t_k\},\, \lambda \times 100) \;=\; \min\Big(\lfloor \tfrac{T\lambda}{100} \rfloor,\, T-1\Big).$$

where $0 \;\leq\; \lambda \;\leq\; 1$. At step t_μ, we average the two denoised estimates,

$$x_{0|t_\mu} = \frac{1}{2}\big(x_{0|t_G} + x_{0|t_{LoG}}\big),$$

and append it as a third sample in the batch as shown in figure 5.4.

Since the "average-face" sample is introduced only after the preconditioned update at t_μ, it has no prior noise residual. To equip it with consistent

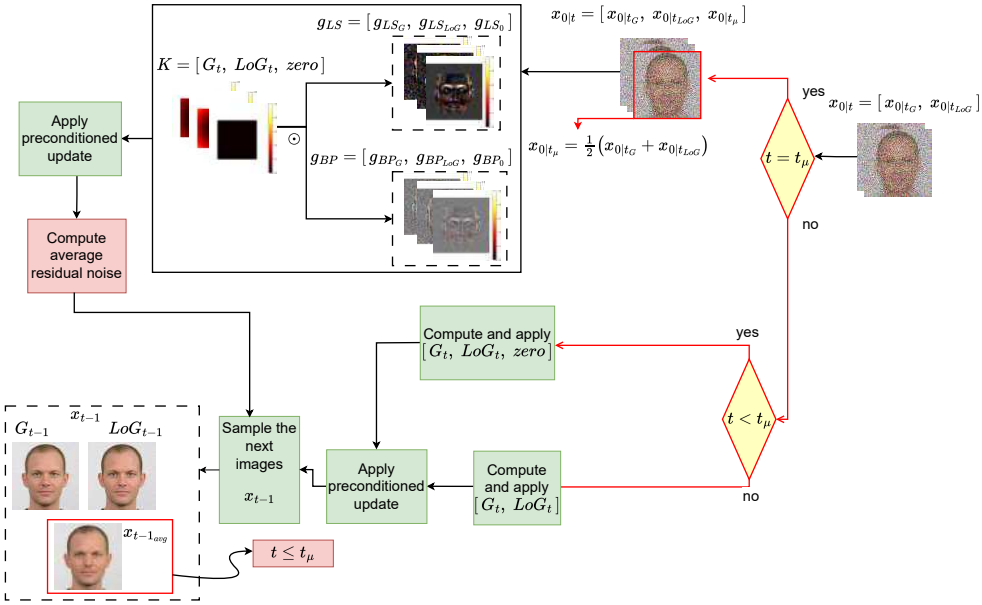


Figure 5.4: Image sampling when the "average-face" inference needs to be generated and processed in the subsequent steps.

stochasticity, we form its noise as the arithmetic mean of the two existing residuals,

$$\hat{\epsilon}_t^{\mu} = \frac{1}{2}\big(\hat{\epsilon}_t^{(1)} + \hat{\epsilon}_t^{(2)}\big),$$

and draw a fresh Gaussian perturbation $\{\epsilon^{(i)}\}_{i=1}^{s} \sim \mathcal{N}(0, I_n)$ where s is the number of samples in the batch. We then apply the usual noise-injection step. This novel averaging of residual noise—previously non-existent for the new branch—ensures that the mean inference at t_μ is a coherent, third sample derived from the Gaussian- and LoG-guided paths, thereby creating a genuinely new deidentified face at that timestep.

Choosing an earlier percentile (big λ which is a percentile at the beginning of the Markov chain for the denoising process, hence big t_μ) injects the average when the chain is still highly noisy. Since fewer denoising steps have been applied, the two samples remain closer to pure Gaussian noise, so their mean is more random—yielding a face that is harder to recognize and thus more privacy-protective. In practice, for $t_\mu \lesssim 500$ (approximately the first

half of a typical $T = 1000$ schedule), $x_{0|t_\mu}$ is dominated by noise, and the average converges to a nearly random face unique to that point in the Markov chain.

By contrast, selecting a late percentile (small λ) injects an average closer to the final reconstruction, preserving more facial detail and offering a milder privacy level. This flexibility allows users to trade off randomness (privacy) versus realism simply by tuning λ.

Chapter 6

Experiments

This chapter presents all the experiments conducted to evaluate both MaskD-DPG and GauDeiDiff. First, we introduce the benchmark datasets and performance metrics used to assess the models. Next, we compare our model's experimental results with those of state-of-the-art methods. Finally, we report the outcomes of ablation studies on our models and additional experiments.

6.1 Datasets

To demonstrate cross-domain generalization, we employ four distinct facial image datasets. Two of these comprise images captured in controlled environments—typically in studios with uniform backgrounds—as shown in Figure 6.1. For the other two datasets, they contain in-the-wild images acquired in unconstrained scenarios with complex backgrounds (Figure 6.2). One of the studio datasets we consider is **The Extended M2VTS database (XM2VTS)** [61]. The XM2VTS corpus is a multi-modal biometric collection comprising high-resolution color photographs, 32 kHz/16-bit audio recordings, video clips, and 3D facial meshes. In this study, we restrict our analysis to the still images: each of the 295 subjects contributes eight color portraits, yielding 2359 images in total. All photographs were acquired under

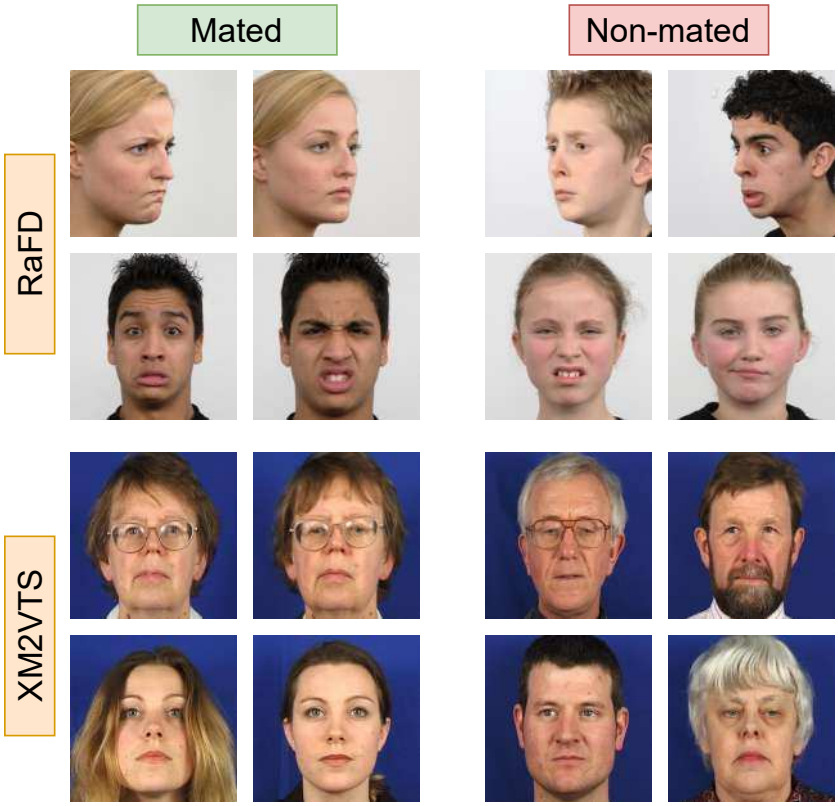


Figure 6.1: Studio image pairs of mated (same identity) and non-mated (different identity) subjects, captured at varying angles and expressions across the selected datasets.

controlled studio conditions—uniform blue backdrops and near-frontal head poses with subjects facing directly at the camera. Since XM2VTS provides only identity labels, our evaluation on this dataset focuses exclusively on face verification. We therefore constructed 1472 genuine (mated) pairs and 1472 impostor (non-mated) pairs. Like the RaFD collection, XM2VTS poses a deidentification challenge: its studio-quality captures accentuate distinctive facial characteristics, making obfuscation more difficult.

Another studio dataset is the **Radboud Faces Database (RaFD)** [62]. It contains high-resolution studio photographs of 73 participants, each por-

traying eight annotated emotional expressions (anger, disgust, fear, happiness, sadness, surprise, contempt, and neutral) against a white background under consistent lighting. The complete dataset offers images from five different camera angles and three distinct gaze directions. For our experiments, we selected images taken at the 45°, 90°, and 135° angles. The subset used in our experiments comprises 1607 images. For the genuine and impostor pairs, we have 1,608 pairs each. For unconstrained settings, we employ the

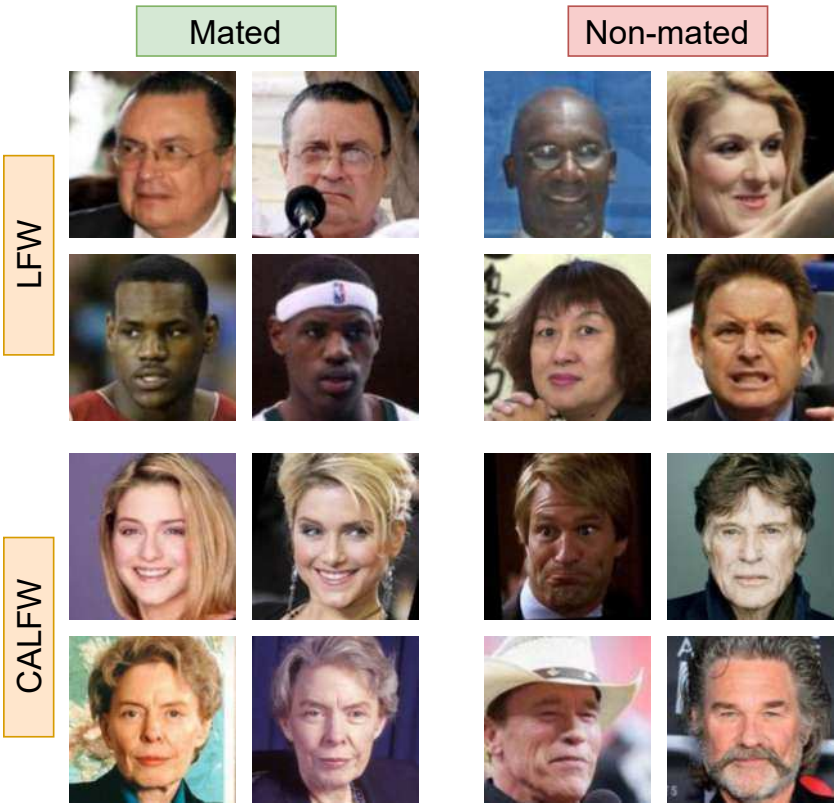


Figure 6.2: "In the wild" image pairs of mated (same identity) and non-mated (different identity) subjects, captured at different backgrounds across the selected datasets.

Labeled Faces in the Wild (LFW) dataset [63], which comprises 13,233 face photographs of 5,749 individuals collected "in the wild" with natural

variability in pose, illumination, expression, and background. To guarantee the formation of genuine pairs, we restrict our experiments to the 1,680 subjects who have at least two images each. From this subset, we randomly sample 3000 genuine (same-identity) pairs and 3000 impostor (different-identity) pairs following standard verification protocols.

Cross-Age LFW (CALFW) [64] extends the original LFW benchmark by deliberately introducing large age gaps into positive pairs while controlling for gender and race in negative pairs. The dataset comprises 4025 identities, each represented by 2–4 studio-aligned images obtained via a similarity transform on the original LFW faces, yielding a total of 11517 images. CALFW follows the standard LFW 10-fold verification protocol: each fold contains 300 genuine (same-identity) pairs chosen to maximize age difference per subject, and 300 impostor (different-identity) pairs drawn randomly from individuals of the same gender and race. Aggregated across all folds, this produces 3000 genuine and 3000 impostor pairs, making CALFW a more challenging test of real-world face verification under cross-age variation.

6.2 Performance Metrics

6.2.1 ID verification metrics

In this work, we evaluate identity verification performance using a series of classification and biometric metrics derived from pairwise cosine similarity scores between embeddings (as shown in equation 4.1). The evaluation is conducted using genuine pairs (same identity) and impostor pairs (different identities). The following metrics are used to comprehensively assess the system's behavior under different de-identification techniques.

Cosine Similarity: given two feature vectors $\mathbf{a}$ and $\mathbf{b}$ representing face embeddings, cosine similarity is defined as:

$$\text{cosine_sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \tag{6.1}$$

This measure ranges from -1 to 1, with 1 indicating perfect similarity and

-1 perfect dissimilarity. In our setting, cosine similarities are normalized to the $[0, 1]$ range and serve as the primary comparison metric for genuine vs. impostor pair distributions.

False Acceptance Rate (FAR) and False Rejection Rate (FRR): these are standard biometric verification metrics. For a given decision threshold τ:

- **FAR** is the proportion of impostor pairs incorrectly classified as mated:

$$\text{FAR}(\tau) = \frac{\#\text{ non-mated pairs with score} \geq \tau}{\text{total }\#\text{ of non-mated pairs}} \tag{6.2}$$

- **FRR** is the proportion of mated pairs incorrectly classified as non-mated:

$$\text{FRR}(\tau) = \frac{\#\text{ mated pairs with score} < \tau}{\text{total }\#\text{ of mated pairs}} \tag{6.3}$$

These metrics are used to construct the Receiver Operating Characteristic (ROC) curve and are also critical in the calculation of Equal Error Rate.

Equal Error Rate (EER): the Equal Error Rate is the value of the threshold τ where FAR and FRR are equal:

$$\text{EER} = \text{FAR}(\tau^*) = \text{FRR}(\tau^*) \tag{6.4}$$

where τ^* is the threshold at which this equality approximately holds. EER provides a single scalar summary of verification performance and is particularly useful for comparing systems independently of operating point.

Area Under the ROC Curve (AUC): AUC quantifies the overall separability of the mated and non-mated score distributions across all thresholds. It is defined as:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x))\, dx \tag{6.5}$$

where $\text{TPR} = 1 - \text{FRR}$ and $\text{FPR} = \text{FAR}$. Higher AUC values (closer to 1) indicate better discrimination between mated and non-mated pairs.

Bootstrapping: to assess the stability and variance of the computed FAR, FRR, AUC, and EER, we employ bootstrapping by resampling the

mated and non-mated score distributions multiple times. This allows us to compute uncertainty estimates (e.g., standard deviations) for each metric, ensuring statistical robustness.

Confusion Matrix and Derived Metrics: at the EER threshold, we classify scores into binary decisions (genuine/impostor) and compute the following metrics from the confusion matrix:

- **Accuracy (Acc):**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6.6}$$

  where TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives.

- **Precision (Prec):**

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6.7}$$

- **Recall (Rec):**

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6.8}$$

- **F1 Score:**

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6.9}$$

- **Matthews Correlation Coefficient (MCC):**

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{6.10}$$

These classification metrics complement the ROC and EER analysis by offering additional insight into the verification system's reliability at a specific operating threshold.

Visualizations: to support our analysis, we visualize:

1. ROC curves with confidence bands derived from bootstrapping.
2. FAR and FRR curves across thresholds with EER lines.

These visual tools help qualitatively assess the behavior of different de-identification techniques under identity verification constraints.

6.2.2 Image quality metrics

To evaluate the fidelity and diversity of generated images, we employ a suite of complementary metrics. All metrics operate on a feature representation of images (e.g. features from DINOv2-ViT-L/14 [65]), except for memorization checks which act in pixel space.

Fréchet Distance (FD): let (μ_r, Σ_r) and (μ_g, Σ_g) be the sample mean and covariance of real and generated features, respectively. The Fréchet distance defined as:

$$\mathrm{FD}(\mu_r, \Sigma_r;\, \mu_g, \Sigma_g) \;=\; \|\mu_r - \mu_g\|_2^2 \;+\; \mathrm{Tr}\big(\Sigma_r + \Sigma_g - 2\,(\Sigma_r \Sigma_g)^{1/2}\big). \tag{6.11}$$

Intuitively, FD measures the 2-Wasserstein distance between the two Gaussians fit to real and generated data in feature space—lower FD indicates closer match in mean and covariance.

Kernel Distance (KD): an alternative metric that does not assume Gaussianity uses the unbiased estimator of the squared Maximum Mean Discrepancy (MMD^2) with kernel k:

$$\begin{aligned}\mathrm{KD}(\{x_g\}, \{x_r\}) = {} & \frac{1}{n(n-1)} \sum_{i \neq i'} k(x_i^g, x_{i'}^g) \\ & + \frac{1}{m(m-1)} \sum_{j \neq j'} k(x_j^r, x_{j'}^r) \\ & - \frac{2}{nm} \sum_{i,j} k(x_i^g, x_j^r)\,.\end{aligned} \tag{6.12}$$

This yields a proper distance between the full distributions of real and generated features; in practice a 3rd-degree polynomial kernel is used. The sampling variance of this U-statistic can be estimated either in closed form or by computing KD over multiple random splits of the data, yielding a "KD variance" that quantifies uncertainty in the estimate.

- $\dfrac{1}{n(n-1)} \displaystyle\sum_{i \neq i'} k(x_i^g, x_{i'}^g)$: the average kernel similarity among all *pairs* of **generated samples**. Dividing by $n(n-1)$ normalizes over the $n(n-1)$

ordered pairs (i, i'). Intuitively, a mode-collapsed generator yields high gen–gen similarity.

- $\frac{1}{m(m-1)} \sum_{j \neq j'} k(x_j^r, x_{j'}^r)$: the average kernel similarity among all *pairs* of **real samples**. This real–real term serves as a baseline capturing the cohesiveness of the real data.

- $\frac{2}{nm} \sum_{i,j} k(x_i^g, x_j^r)$: twice the average cross-kernel similarity between generated and real samples. Large gen–real similarity indicates that generated points lie close to real data in feature space.

When the generated and real distributions coincide, the gen–gen and real–real terms both match the gen–real term, yielding KD = 0. Any discrepancy in intra-set or cross-set similarity makes KD > 0.

Precision and Recall: following Sajjadi et al. [66], we compute precision and recall via nearest-neighbour occupancy in feature space. Let

$$S(\{x\}) = \bigcup_{x \in \{x\}} B\big(x, \mathrm{NND}_k(x)\big)$$

be the union of Euclidean balls around each reference sample of radius equal to its k-th nearest-neighbour distance (NND). Then

$$\text{precision} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\big(x_i^g \in S(\{x_j^r\}_{j=1}^m)\big), \tag{6.13}$$

$$\text{recall} = \frac{1}{m} \sum_{j=1}^{m} \mathbf{1}\big(x_j^r \in S(\{x_i^g\}_{i=1}^n)\big). \tag{6.14}$$

Precision measures the fraction of generated samples that fall into high-density regions of the real data; recall measures the fraction of real-data regions that are covered by the generator.

Density and Coverage: also based on the same neighbourhood balls,

Naeem et al. [67] proposed:

$$\text{density} = \frac{1}{k\,n}\sum_{i=1}^{n}\sum_{j=1}^{m}\mathbf{1}\big(x_i^g \in B(x_j^r, \mathrm{NND}_k(x_j^r))\big), \tag{6.15}$$

$$\text{coverage} = \frac{1}{m}\sum_{j=1}^{m}\mathbf{1}\ \exists\ i\ s.t\ \big(x_i^g \in B(x_j^r, \mathrm{NND}_k(x_j^r))\big). \tag{6.16}$$

Density counts, on average, how many real-data neighbourhoods each generated sample lies in; coverage measures the fraction of real-data neighbourhoods that contain at least one generated sample.

Overall, FD and KD both assess how closely the generated distribution matches the real distribution in feature space—FD does so by comparing Gaussian moments (means and covariances), while KD uses a kernel-based MMD to capture more general differences. Precision and recall then break this comparison into two complementary aspects: precision quantifies fidelity by measuring the fraction of generated samples that lie within high-density regions of the real data, and recall quantifies diversity by measuring the fraction of real-data regions that the generator covers. Density and coverage offer softened versions of these diagnostics—density acts like a "soft" precision by counting how many real-data neighborhoods each generated sample falls into on average, and coverage acts like a "soft" recall by measuring the fraction of real-data neighborhoods that contain at least one generated sample. Finally, by reporting not only the KD estimate but also its sampling variance (computed via U-statistic theory or over random data splits), we can judge whether observed differences are statistically significant.

Mean Squared Error (MSE): let $\{I_j^r\}_{j=1}^m$ and $\{I_i^g\}_{i=1}^n$ be the real and generated images (in pixel space), each of size $H \times W \times C$. The mean squared error is

$$\mathrm{MSE} \;=\; \frac{1}{mnHC}\sum_{i=1}^{n}\sum_{j=1}^{m}\big\|I_i^g - I_j^r\big\|_2^2,$$

or more commonly when paired sample-wise,

$$\mathrm{MSE} \;=\; \frac{1}{nHC}\sum_{i=1}^{n}\big\|I_i^g - I_i^r\big\|_2^2.$$

MSE directly measures average per-pixel squared deviation between generated and real images, exposing low-level artifacts such as blurring, color shifts, or noise patterns that feature-space distances may overlook. Including pixel-level MSE is important in the thesis because it provides a complementary perspective on fidelity: while FD, KD, precision/recall, density, and coverage all operate on learned feature representations and capture global or semantic alignment, MSE anchors evaluation in raw pixel consistency. This ensures that our models not only match the statistical properties of features but also faithfully reproduce fine-grained visual details.

6.3 Experimental Results

6.3.1 Verification Results

We explore deidentification performance based on verification strength using AdaFace [28]. This is a state-of-the-art recognition model that was trained on the large-scale MS1MV2 face dataset with a margin-based loss function for face recognition, which adaptively adjusts its angular and additive margins based on an image's quality, approximated by the feature norm. This recognition model uses a ResNet backbone to extract 512-dimensional embeddings.

We also compare our performance against seven deidentification approaches from the literature review, including GAN-based methods (DeepPrivacy [36], CPP-DeID [31], AMT-GAN [21]), VAE-based approach (CLEANIR [16]) and diffusion-based methods (FAMS [47], LDFA [44], FADM [49]).

In the first part of the experiments, we evaluate the strength of our methods in controlled environments such as the RaFD [62] and XM2VTS [61] datasets; this can be observed in Figure 6.3. For the RaFD dataset, Figure 6.3a shows how our approaches, MaskDDPG and GDDiff, produces ROC curves across all 100 thresholds used for mated and non-mated image pairs similarities based on standardized cosine similarity. As the parameter λ increases, our approaches reduces the area under the curve (AUC), reaching a

minimum AUC of 0.84 for MaskDDPG $\lambda = 1.0$ and 0.77 for Mean-GDDiff $\lambda = 0.5$, which represents the maximum deidentification achieved in this setting for our methods. In Figure 6.3b, we compare this maximum deidentification level against all other competitors. CPP-DeID [31] and CLEANIR [16] are the two approaches that outperform MaskDDPG by achieving AUC values below 0.84. Achieving deidentification in these studio settings is particularly challenging because the images are captured under consistent lighting, pose, and background conditions, resulting in high-quality, low-variance data; consequently, the verification model is exceptionally accurate at finding similarities between image pairs, making it harder to obscure identity. For the case of Mean-GDDiff $\lambda = 0.5$ we can observe that ranks second in the list; because the kernels obfuscate more area of the face (like hair, ears, etc.) that MaskDDPG does not obfuscate. For the case of the Laplacian (lpl) and Gaussian (gauss) results also appear to behave superficially in terms of ID removal hence we are only going to focus on the analysis of the performance of the mean inference generated from these two trajectories at different λ values which indicate how early we trigger this average inference in the Markov process.

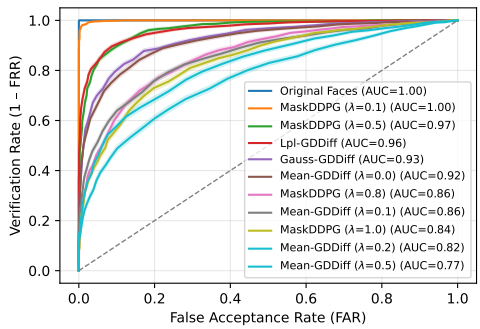
In the second part of these experiments on studio images, we observe a similar pattern with the XM2VTS [61] dataset, where MaskDDPG and GDDiFF also apply deidentification across different λ values in figure 6.3c. Our minimum AUC on this dataset is 0.90 for MaskDDPG and 0.69 for GDDiff. MaskDDPG compared to our competitors (Figure 6.3d), we rank fifth behind FAMS [47], LDFA [44], and CPP-DeID [31]. For GDDiff, we ranked as the best model for deidentification in this dataset outperforming even CPP-DeID. Almost all AUC values for all approaches here are higher than those achieved on RaFD, likely because RaFD contains greater variability in expressions and poses, whereas XM2VTS consists of mostly constant, frontal facial images. In both cases, however, the verification model remains highly accurate at detecting similarities between image pairs, preventing any approach from achieving AUC values below 0.69 in XM2VTS

dataset. GDDiff is the only model that achieves AUC performance lower than 0.70, which is the only method performing that well in ID removal. It is important to note that some competitors and GDDiff alter the background or modify regions beyond the face—such as hair and clothing—as done by DeepPrivacy [36] and FADM [49]. This additional context alteration can lead to lower AUC values since more of the image is changed as shown with GDDiff results. Despite XM2VTS's low variability, our model MaskDDPG—by altering only the facial region—still outperforms these before-mentioned methods, demonstrating stronger identity removal even when only the face is modified.

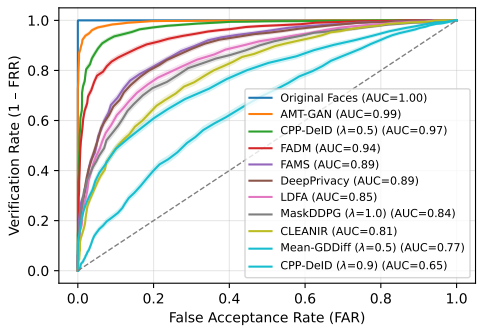
Differently from studio-based datasets, images "in the wild" tend to exhibit more variability—different backgrounds, poses, and occlusions; so the verification model struggles more to find similarities between images. Although this setting better approximates real-world scenarios, where one must protect the subject's identity while preserving non-identifying features and background context, it is more challenging to balance these requirements. We conducted the same verification experiments on the LFW dataset [63], which contains images in the wild, and observed that our approaches also produce different ROC curves with decreasing AUC as λ increases (Fig. 6.4a). Notably, our model MaskDDPG performs similarly for $\lambda = 0.8$ and $\lambda = 1.0$, so we use $\lambda = 1.0$ when comparing against competitors in Figure 6.4b. Similarly for Mean-GDDiff $\lambda = 0.5$.

In the LFW experiment, our MaskDDPG model ranks sixth, with AUC values very similar to CLEANIR [16], FAMS [47], and LDFA [44], which occupy higher positions. For our second approach, Mean-GDDiff ($\lambda = 0.5$), it ranks second behind CPP-DeID [31], which, when using its privacy protection parameter at 0.9, achieves an AUC of 0.60; close to random-guess behavior.

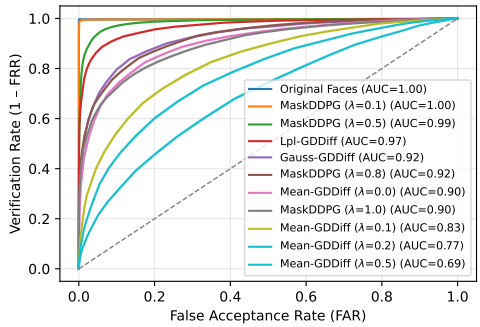
The CALFW dataset introduces additional complexity because its mated and non-mated pairs have greater age differences. Consequently, the original images (without deidentification) already yields an AUC less than 1, as shown in Figure 6.4c. In this dataset, our method MaskDDPG ranks fifth in terms of AUC but remains comparable to CLEANIR [16], FAMS [47], and



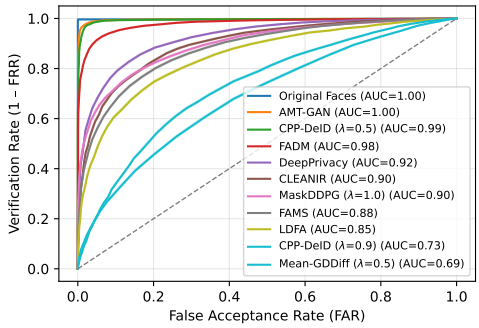
(a) Our methods on RaFD



(b) All methods on RaFD



(c) Our methods on XM2VTS

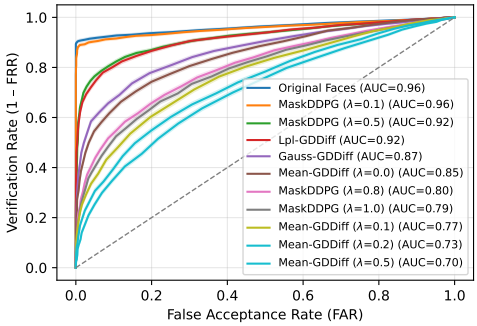


(d) All methods on XM2VTS

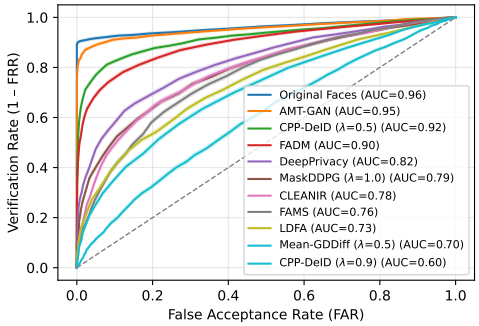
Figure 6.3: ROC curves for the studio datasets: RaFD (top row) and XM2VTS (bottom row).

LDFA [44] (Fig. 6.4d). This suggests that, in terms of identity removal, these approaches perform similarly; therefore, deidentification quality may not be fully captured by AUC alone, and other evaluation metrics may be necessary to distinguish them. Also, in this dataset as well, our method mean GDDiff and CPP-DeID separate from the rest of the approaches as the best deidentifiers in terms of AUC from the plotted ROC curves.

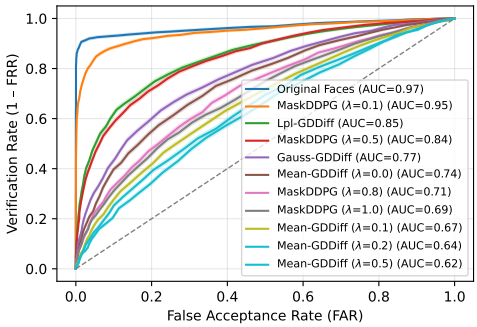
For this reason, we decided to first obtain the threshold τ where the false acceptance rate (FAR) and the false rejection rate (FRR) are the same. This is important because it corresponds to the equal error rate (EER) operating point, which provides a balanced measure of the system's performance by minimizing both security risks (false acceptances) and usability issues in the



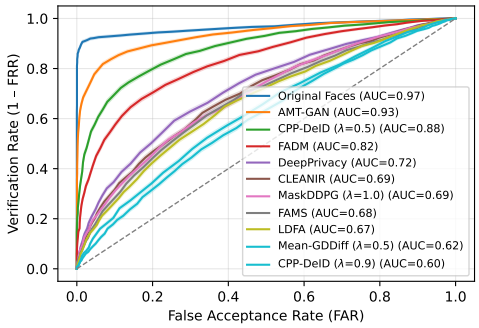
(a) Our methods on LFW



(b) All methods on LFW



(c) Our methods on CALFW



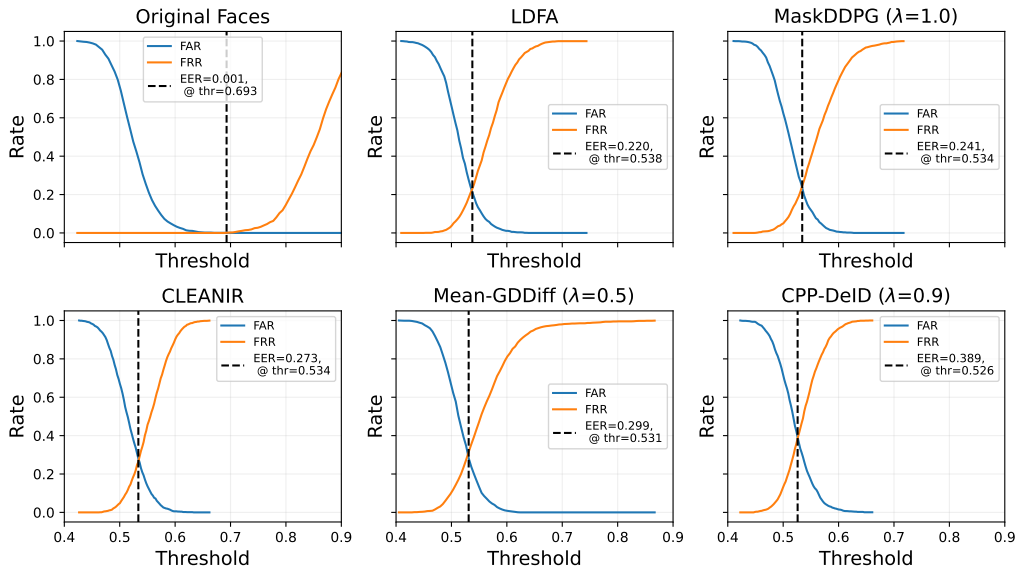
(d) All methods on CALFW

Figure 6.4: ROC curves for the in-the-wild datasets: LFW (top row) and CALFW (bottom row).

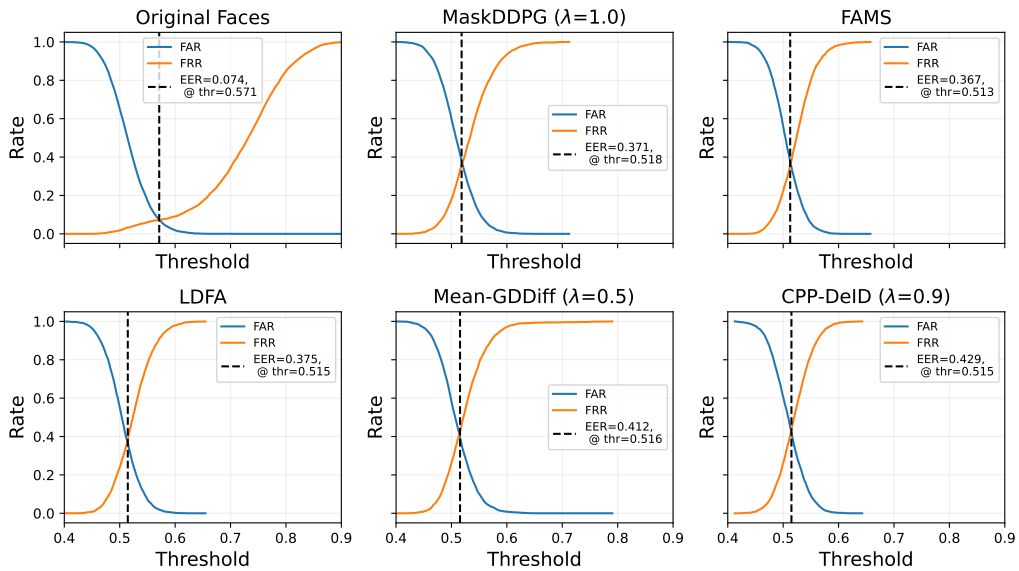
system (false rejections). By evaluating our deidentification methods at τ, we ensure that comparisons against other techniques are made under a neutral operating condition, where neither error type is disproportionately favored.

Moreover, the EER threshold offers a standardized metric for assessing verification accuracy, enabling a clearer understanding of how deidentification impacts identification reliability and user convenience.

This can be observed in Figure 6.5, where, for the case of RaFD (Fig. 6.5b), CPP-DeID had the lowest threshold among all. For the case of CALFW, one can observe an important phenomenon (Fig. 6.5a), where our method MaskDDPG ended up in fifth place, having one of the top 5 τ values. In terms of ERR for CALFW, we outperform CLEANIR where in AUC it was



(a) RaFD



(b) CALFW

Figure 6.5: False Acceptance / False Rejection rate curves for all four top techniques based in 100 disjoint thresholds. ERR threshold shows where these rates are equal.

the opposite, which illustrates a difference in evaluation between these two methods. This indicates that if we choose each EER threshold to classify mated and non-mated pairs, the classification results will differ for each evaluation method.

Once we computed all the thresholds τ per technique, we can show the detailed classification metrics at EER for each experiment. In the RaFD group (Table 6.1), the original (unmodified) images yield an F1 score of 0.999, indicating almost perfect verification. As λ increases for MaskDDPG, the F1 score decreases from 0.983 at $\lambda = 0.1$ to 0.758 at $\lambda = 1.0$, demonstrating progressively stronger deidentification. In parallel, GDDiFF trajectories also exhibit significant degradation in verification performance: the Laplacian trajectory (Lpl-GDDiff) achieves an F1 of 0.904 at its EER threshold (thr = 0.567), and the Gaussian trajectory (Gauss-GDDiff) reaches 0.849 (thr = 0.553). Using the Mean-GDDiff strategy, the average ($\lambda = 0.0$) yields F1 = 0.842, which further decreases to 0.782 at $\lambda = 0.1$, 0.746 at $\lambda = 0.2$, and 0.697 at $\lambda = 0.5$. Thus, at moderate obfuscation levels (e.g., Mean-GDDiff $\lambda = 0.2$), GDDiFF already outperforms most competitors (e.g., LDFA F1 = 0.778, CLEANIR F1 = 0.730) while applying more flexible, diffusion-based obfuscation. At its most aggressive setting (Mean-GDDiff $\lambda = 0.5$), GDDiFF rivals CPP-DeID ($\lambda = 0.9$) by pushing the F1 down to 0.697, indicating that GDDiFF can achieve stronger deidentification than MaskDDPG at lower λ values, although with broader area alterations due to the diffusion process.

In the XM2VTS group (Table 6.2), the original F1 score is 0.996, with CPP-DeID ($\lambda = 0.0$) close behind at 0.995. MaskDDPG at $\lambda = 0.1$ yields an F1 of 0.993, which drops to 0.945 at $\lambda = 0.5$ and 0.812 at $\lambda = 1.0$. GDDiFF again shows a graceful trade-off: Lpl-GDDiff achieves F1 = 0.911 (thr = 0.561) and Gauss-GDDiff records 0.850 (thr = 0.549), highlighting moderate deidentification while preserving more facial structure. Mean-GDDiff at $\lambda = 0.0$ yields F1 = 0.825; as λ increases, it reduces to 0.754 ($\lambda = 0.1$), 0.710 ($\lambda = 0.2$), and 0.624 ($\lambda = 0.5$). Notably, at $\lambda = 0.5$, Mean-GDDiff's F1 of 0.624 outperforms all MaskDDPG settings including CPP-DeID ($\lambda = 0.9$) in

Table 6.1: Summary of metrics at the Equal Error Rate (EER) on RaFD, sorted by F1 score. Experiments use studio images with controlled environments. CPP-DeID ($\lambda = 0.9$) and Mean-GDDiff ($\lambda = 0.5$) emerge as the top performers; here `thr` denotes the EER threshold, `Acc` the accuracy (↓), `Prec` the precision (↓), `Rec` the recall (↓), and `F1` the F1 score (↓).

Method	thr	Acc ↓	Prec ↓	Rec ↓	F1 ↓
Original	0.693	0.999	0.999	0.999	0.999
CPP-DeID ($\lambda = 0.0$)	0.606	0.984	0.984	0.983	0.984
MaskDDPG ($\lambda = 0.1$)	0.612	0.983	0.981	0.985	0.983
CPP-DeID ($\lambda = 0.2$)	0.594	0.979	0.978	0.979	0.979
AMT-GAN	0.584	0.962	0.963	0.960	0.962
CPP-DeID ($\lambda = 0.5$)	0.566	0.922	0.920	0.924	0.922
Lpl-GDDiff	0.567	0.905	0.912	0.896	0.904
MaskDDPG ($\lambda = 0.5$)	0.565	0.899	0.899	0.898	0.898
FADM	0.559	0.869	0.869	0.869	0.869
Gauss-GDDiff	0.553	0.850	0.854	0.844	0.849
Mean-GDDiff (λ=0.0)	0.552	0.842	0.841	0.842	0.842
FAMS	0.546	0.809	0.810	0.806	0.808
DeepPrivacy	0.548	0.806	0.811	0.798	0.805
Mean-GDDiff (λ=0.1)	0.540	0.778	0.770	0.794	0.782
MaskDDPG ($\lambda = 0.8$)	0.537	0.781	0.786	0.771	0.779
LDFA	0.538	0.780	0.785	0.770	0.778
MaskDDPG ($\lambda = 1.0$)	0.534	0.759	0.761	0.754	0.758
Mean-GDDiff (λ=0.2)	0.532	0.740	0.730	0.763	0.746
CLEANIR	0.534	0.728	0.724	0.736	0.730
Mean-GDDiff (λ=0.5)	0.531	0.702	0.708	0.686	0.697
CPP-DeID ($\lambda = 0.9$)	0.526	0.611	0.611	0.614	0.612

terms of lowering verification accuracy, demonstrating that diffusion-based averaging can be more effective than mask-based obfuscation in studio settings. However, this comes with more extensive image alterations compared to the localized occlusions of MaskDDPG. In contrast, at $\lambda = 1.0$, MaskDDPG's F1 of 0.812 still outperforms DeepPrivacy (0.847) and CLEANIR (0.817) while altering only the face.

Table 6.2: Summary of metrics at the Equal Error Rate (EER) on XM2VTS, sorted by F1 score. Experiments use studio images with controlled environments. Mean-GDDiff ($\lambda = 0.5$) and CPP-DeID ($\lambda = 0.9$) emerge as the top performers.

Method	thr	Acc ↓	Prec ↓	Rec ↓	F1 ↓
Original	0.644	0.996	0.996	0.996	0.996
CPP-DeID ($\lambda = 0.0$)	0.624	0.995	0.995	0.995	0.995
CPP-DeID ($\lambda = 0.2$)	0.616	0.993	0.993	0.994	0.993
MaskDDPG ($\lambda = 0.1$)	0.620	0.993	0.992	0.993	0.993
AMT-GAN	0.584	0.978	0.977	0.980	0.978
CPP-DeID ($\lambda = 0.5$)	0.584	0.974	0.976	0.973	0.974
MaskDDPG ($\lambda = 0.5$)	0.573	0.945	0.944	0.947	0.945
FADM	0.568	0.932	0.936	0.928	0.932
Lpl-GDDiff	0.561	0.911	0.913	0.909	0.911
Gauss-GDDiff	0.549	0.852	0.862	0.837	0.850
DeepPrivacy	0.545	0.846	0.838	0.856	0.847
MaskDDPG ($\lambda = 0.8$)	0.548	0.842	0.842	0.842	0.842
Mean-GDDiff (λ=0.0)	0.544	0.825	0.825	0.825	0.825
CLEANIR	0.540	0.816	0.815	0.818	0.817
MaskDDPG ($\lambda = 1.0$)	0.546	0.814	0.824	0.800	0.812
FAMS	0.538	0.798	0.796	0.802	0.799
LDFA	0.536	0.769	0.766	0.776	0.771
Mean-GDDiff (λ=0.1)	0.531	0.747	0.733	0.775	0.754
Mean-GDDiff (λ=0.2)	0.526	0.701	0.690	0.732	0.710
CPP-DeID ($\lambda = 0.9$)	0.530	0.675	0.678	0.667	0.672
Mean-GDDiff (λ=0.5)	0.523	0.636	0.645	0.605	0.624

For the "in-the-wild" datasets (LFW and CALFW, Tables 6.3 and 6.4),

both MaskDDPG and GDDiFF follow analogous trends. On LFW, MaskDDPG's F1 decreases from 0.917 ($\lambda = 0.1$) to 0.708 ($\lambda = 1.0$), whereas GDDiFF's Gauss trajectory drops to 0.788 (thr = 0.527) and Mean-GDDiff at $\lambda = 0.5$ reaches 0.646. In CALFW, MaskDDPG at $\lambda = 1.0$ yields 0.633, while Gauss-GDDiff and Mean-GDDiff at $\lambda = 0.5$ achieve 0.699 and 0.581, respectively. These results confirm that kernel methods as GDDiff methods can match or exceed MaskDDPG's performance at comparable obfuscation levels, offering a tunable mechanism for balancing identity removal and image realism.

Table 6.3: Summary of metrics at the Equal Error Rate (EER) on LFW, sorted by F1 score. CPP-DeID ($\lambda = 0.9$) and Mean-GDDiff ($\lambda = 0.5$) emerge as the top performers.

Method	thr	Acc ↓	Prec ↓	Rec ↓	F1 ↓
Original	0.544	0.922	0.922	0.922	0.922
MaskDDPG ($\lambda = 0.1$)	0.544	0.917	0.921	0.913	0.917
CPP-DeID ($\lambda = 0.0$)	0.542	0.915	0.921	0.908	0.915
AMT-GAN	0.538	0.908	0.906	0.910	0.908
CPP-DeID ($\lambda = 0.2$)	0.537	0.899	0.896	0.903	0.899
CPP-DeID ($\lambda = 0.5$)	0.533	0.853	0.853	0.853	0.853
MaskDDPG ($\lambda = 0.5$)	0.533	0.849	0.843	0.858	0.850
Lpl-GDDiff	0.532	0.843	0.842	0.845	0.844
FADM	0.530	0.823	0.832	0.809	0.820
Gauss-GDDiff	0.527	0.791	0.800	0.777	0.788
Mean-GDDiff (λ=0.0)	0.523	0.763	0.758	0.774	0.766
DeepPrivacy	0.519	0.743	0.737	0.754	0.745
MaskDDPG ($\lambda = 0.8$)	0.520	0.723	0.723	0.724	0.723
CLEANIR	0.518	0.714	0.719	0.705	0.712
MaskDDPG ($\lambda = 1.0$)	0.521	0.713	0.721	0.696	0.708
Mean-GDDiff (λ=0.1)	0.518	0.699	0.695	0.711	0.703
FAMS	0.514	0.696	0.696	0.698	0.697
LDFA	0.514	0.670	0.666	0.679	0.673
Mean-GDDiff (λ=0.2)	0.518	0.673	0.680	0.652	0.666
Mean-GDDiff (λ=0.5)	0.515	0.651	0.656	0.636	0.646
CPP-DeID ($\lambda = 0.9$)	0.510	0.571	0.571	0.573	0.572

Table 6.4: Summary of metrics at the Equal Error Rate (EER) on CALFW, sorted by F1 score. CPP-DeID ($\lambda = 0.9$) and Mean-GDDiff ($\lambda = 0.5$) emerge as the top performers.

Method	thr	Acc ↓	Prec ↓	Rec ↓	F1 ↓
Original	0.571	0.926	0.925	0.927	0.926
MaskDDPG ($\lambda = 0.1$)	0.559	0.890	0.890	0.890	0.890
CPP-DeID ($\lambda = 0.0$)	0.554	0.887	0.885	0.888	0.887
CPP-DeID ($\lambda = 0.2$)	0.549	0.868	0.862	0.876	0.869
AMT-GAN	0.547	0.868	0.867	0.868	0.868
CPP-DeID ($\lambda = 0.5$)	0.539	0.800	0.804	0.793	0.798
Lpl-GDDiff	0.534	0.766	0.759	0.777	0.768
MaskDDPG ($\lambda = 0.5$)	0.536	0.759	0.758	0.760	0.759
FADM	0.533	0.745	0.738	0.759	0.748
Gauss-GDDiff	0.528	0.702	0.704	0.695	0.699
Mean-GDDiff (λ=0.0)	0.524	0.678	0.673	0.690	0.681
DeepPrivacy	0.519	0.659	0.657	0.663	0.660
MaskDDPG ($\lambda = 0.8$)	0.519	0.652	0.648	0.663	0.655
CLEANIR	0.513	0.644	0.642	0.647	0.645
FAMS	0.513	0.633	0.629	0.642	0.636
MaskDDPG ($\lambda = 1.0$)	0.518	0.629	0.625	0.642	0.633
Mean-GDDiff (λ=0.1)	0.518	0.618	0.614	0.630	0.622
LDFA	0.515	0.625	0.627	0.614	0.621
Mean-GDDiff (λ=0.2)	0.516	0.598	0.595	0.608	0.601
Mean-GDDiff (λ=0.5)	0.516	0.588	0.590	0.573	0.581
CPP-DeID ($\lambda = 0.9$)	0.515	0.570	0.570	0.562	0.566

6.3.2 Image Quality Results

We assess image quality using the metrics proposed by Stein et al. [68], namely Fréchet Distance (FD), kernel distance (KD), precision P, recall R, density D, and coverage C; all these metrics use DinoV2 embeddings [65]. FD measures the discrepancy between the distribution of deidentified images and that of the original images in feature space, thereby capturing perceptual realism. KD serves as an auxiliary divergence measure in the same embedding

space. Precision and recall quantify whether generated samples lie within the support of the original distribution and whether the original distribution is well covered by generated samples, respectively. Density and coverage further decompose recall into measures of mass (density) and extent (coverage) [68]. Table 6.5 reports these metrics for RaFD, Table 6.6 for XM2VTS, Table 6.7 for LFW, and Table 6.8 for CALFW. In addition, Figure 6.6 summarizes the mean squared error (MSE) per dataset and technique, which tracks pixel-level fidelity to the original images.

In both RaFD and XM2VTS, backgrounds, lighting, and poses are tightly controlled, so DinoV2 embeddings primarily encode facial characteristics. Consequently, metrics that depend on diversity of non-facial content—namely P, R, D, and C—report values that are nearly zero for most techniques, as shown in Tables 6.5 and 6.6. For example, in RaFD (Table 6.5), almost all methods yield $R = 0$ and $C = 0$, with $P \leq 0.10$ and $D \leq 0.26$. Our techniques at their strongest settings still show low R and C: MaskDDPG ($\lambda = 1.0$) reports $P = 0.05$, $R = 0.02$, $D = 0.16$, $C = 0.01$, while Mean-GDDiff ($\lambda = 0.5$) has $P = 0.01$, $R = 0.20$, $D = 0.00$, $C = 0.01$. Gauss-GDDiff and Lpl-GDDiff record similarly minimal recall and coverage (e.g., Gauss-GDDiff: $P = 0.02$, $R = 0.25$, $D = 0.00$, $C = 0.02$; Lpl-GDDiff: $P = 0.03$, $R = 0.20$, $D = 0.01$, $C = 0.03$). These near-zero values indicate that changing only the facial region leaves little variation for the DinoV2 model to detect beyond the face, since the background is identical across the dataset. In such settings, FD becomes the most informative realism metric, as it captures distributional shifts in the face embedding space even when non-facial diversity is absent. Indeed, in RaFD (Table 6.5), FD ranges from 422.05 for MaskDDPG ($\lambda = 0.1$) down to 511.59 for MaskDDPG ($\lambda = 0.5$), and up to 1660.59 for CLEANIR. Mean-GDDiff trajectories lie between these extremes: Mean-GDDiff ($\lambda = 0.0$) records FD = 626.94, Mean-GDDiff ($\lambda = 0.1$) FD = 670.18, Mean-GDDiff ($\lambda = 0.2$) FD = 700.49, and Mean-GDDiff ($\lambda = 0.5$) FD = 747.91. Gauss-GDDiff (FD = 625.97) and Lpl-GDDiff (FD = 556.08) fall slightly below the weakest MaskDDPG.

Kernel distance (KD) trends alongside FD (e.g., MaskDDPG ($\lambda = 1.0$) KD = 3.49, Gauss-GDDiff KD = 3.37, Lpl-GDDiff KD = 3.16), corroborating that FD reliably encodes realism when background information is constant.

A similar pattern emerges on XM2VTS (Table 6.6). Gauss-GDDiff (FD = 539.45, KD = 4.26) and Lpl-GDDiff (FD = 457.76, KD = 3.57) achieve moderate FD values, indicating that diffusion-based perturbations produce faces that remain closer to the original manifold than the most aggressive baselines (e.g., CPP-DeID ($\lambda = 0.9$) FD = 943.39, CLEANIR FD = 1226.19). In contrast, DeepPrivacy (FD = 374.91, KD = 3.43) and MaskDDPG ($\lambda = 0.1$) (FD = 258.64, KD = 2.05) achieve the lowest FD—reflecting nearly photo-realistic outputs—at the expense of weaker identity masking. Overall, FD and KD metrics confirm that GDDiFF variants (Mean-, Gauss-, Lpl-) provide a tunable spectrum of realism, with FD increasing monotonically as λ grows, while MaskDDPG remains the most conservative face-only method.

In LFW (Table 6.7), images exhibit substantial variation in background, pose, and occlusions. Consequently, density and coverage become meaningful indicators of whether deidentified images adequately preserve the overall content manifold. Techniques that alter only the face while leaving contextual elements intact yield high density and coverage: MaskDDPG ($\lambda = 0.1$) achieves $P = 0.99$, $R = 0.99$, $D = 0.91$, $C = 0.99$, and Mean-GDDiff ($\lambda = 0.0$) shows $P = 0.94$, $R = 0.93$, $D = 0.77$, $C = 0.90$. Gauss-GDDiff further improves these values: $P = 0.94$, $R = 0.92$, $D = 0.75$, $C = 0.89$. Lpl-GDDiff yields $P = 0.96$, $R = 0.96$, $D = 0.81$, $C = 0.95$. By contrast, CPP-DeID at $\lambda = 0.9$ attains $P = 0.91$, $R = 0.00$, $D = 0.61$, $C = 0.04$, reflecting aggressive face changes that shift images outside the support of the original distribution. FD values in LFW further corroborate these observations: MaskDDPG ($\lambda = 1.0$) records FD = 209.32, DeepPrivacy yields FD = 111.86, Mean-GDDiff ($\lambda = 0.5$) FD = 362.03, and CPP-DeID ($\lambda = 0.9$) has FD = 2121.45. KD follows a similar ranking. Thus, FD successfully captures the reduced realism of strongly deidentified images, while D and C quantify how much of the original distribution's support is preserved by each method.

Table 6.5: Performance metrics on the RaFD dataset, sorted in decreasing order by the Fréchet Distance (FD).

Technique	**FD** ↓	KD ↓	P ↑	R ↑	D ↑	C ↑
CLEANIR	1660.59	14.94	0.07	0.00	0.00	0.00
CPP-DeID (0.9)	1476.96	10.45	0.10	0.00	0.00	0.00
LDFA	998.28	5.92	0.06	0.00	0.09	0.00
AMT-GAN	985.86	8.06	0.05	0.00	0.00	0.00
CPP-DeID (0.5)	900.92	6.24	0.06	0.00	0.01	0.00
CPP-DeID (0.2)	885.81	6.14	0.07	0.00	0.04	0.00
CPP-DeID (0.0)	876.91	6.01	0.06	0.00	0.04	0.00
Mean-GDDiff (0.5)	747.91	3.67	0.01	0.20	0.00	0.01
Mean-GDDiff (0.2)	700.49	3.57	0.01	0.21	0.00	0.01
MaskDDPG (1.0)	675.73	3.49	0.05	0.02	0.16	0.01
Mean-GDDiff (0.1)	670.18	3.51	0.02	0.23	0.00	0.02
MaskDDPG (0.8)	659.82	3.46	0.05	0.01	0.19	0.01
Mean-GDDiff (0.0)	626.94	3.40	0.01	0.22	0.00	0.02
Gauss-GDDiff	625.97	3.37	0.02	0.25	0.00	0.02
Lpl-GDDiff	556.08	3.16	0.03	0.20	0.01	0.03
FAMS	526.76	3.27	0.08	0.24	0.62	0.23
FADM	521.99	2.37	0.05	0.08	0.51	0.08
MaskDDPG (0.5)	511.59	2.96	0.04	0.04	0.20	0.04
DeepPrivacy	448.67	2.37	0.04	0.34	0.18	0.20
MaskDDPG (0.1)	422.05	2.61	0.03	0.09	0.26	0.09

Table 6.6: Performance metrics on the XM2VTS dataset, sorted in decreasing order by the Fréchet Distance (FD).

Technique	**FD** ↓	KD ↓	P ↑	R ↑	D ↑	C ↑
CLEANIR	1226.19	14.39	0.02	0.00	0.00	0.01
AMT-GAN	1174.73	12.50	0.21	0.00	0.07	0.01
CPP-DeID (0.2)	1050.61	11.29	0.01	0.01	0.00	0.00
CPP-DeID (0.0)	1049.91	11.28	0.01	0.00	0.00	0.00
CPP-DeID (0.5)	977.95	10.65	0.01	0.00	0.00	0.00
CPP-DeID (0.9)	943.39	9.65	0.00	0.00	0.00	0.00
LDFA	892.41	6.63	0.05	0.31	0.01	0.04
Mean-GDDiff (0.5)	729.87	5.79	0.04	0.16	0.01	0.02
Mean-GDDiff (0.2)	626.18	4.82	0.05	0.18	0.01	0.03
FAMS	600.45	5.58	0.20	0.44	0.06	0.14
Mean-GDDiff (0.1)	577.48	4.49	0.06	0.20	0.02	0.04
Gauss-GDDiff	539.45	4.26	0.08	0.24	0.02	0.05
Mean-GDDiff (0.0)	526.23	4.15	0.08	0.22	0.02	0.04
FADM	474.29	3.72	0.20	0.56	0.06	0.12
Lpl-GDDiff	457.76	3.57	0.13	0.32	0.04	0.09
MaskDDPG (1.0)	440.57	3.25	0.14	0.50	0.04	0.10
MaskDDPG (0.8)	409.50	3.07	0.16	0.53	0.04	0.11
DeepPrivacy	374.91	3.43	0.33	0.61	0.10	0.26
MaskDDPG (0.5)	324.50	2.52	0.25	0.68	0.07	0.20
MaskDDPG (0.1)	258.64	2.05	0.43	0.80	0.14	0.39

A similar pattern is observed in CALFW (Table 6.8). Deidentified images that maintain realistic backgrounds and facial context—such as MaskDDPG ($\lambda = 1.0$) ($P = 0.91$, $R = 0.90$, $D = 0.72$, $C = 0.91$, FD = 160.41) and Lpl-GDDiff (FD = 168.56, $P = 0.93$, $R = 0.91$, $D = 0.76$, $C = 0.93$)—exhibit both high density/coverage and low FD while being referents in ID protection. Mean-GDDiff ($\lambda = 0.0$) records FD = 192.51, $P = 0.92$, $R = 0.88$, $D = 0.73$, $C = 0.90$. Gauss-GDDiff achieves FD = 194.99, $P = 0.92$, $R = 0.87$, $D = 0.72$, $C = 0.89$. By contrast, CPP-DeID ($\lambda = 0.9$) has poor density ($D = 0.71$) and coverage ($C = 0.07$) with FD = 1615.58, indicating that it produces faces that are perceptually dissimilar across many contexts. FAMS ($P = 0.66$, $R = 0.77$, $D = 0.28$, $C = 0.55$, FD = 548.03) and LDFA ($P = 0.56$, $R = 0.68$, $D = 0.20$, $C = 0.51$, FD = 568.54) perform non-facial context modifications to some extent, which improves coverage relative to face-only methods but still falls short of photo-realism benchmarks, as indicated by intermediate FD and KD values.

Figure 6.6 plots, for each dataset, how different deidentification techniques compare in terms of pixel-level error. In all four cases, the techniques can be ordered from highest to lowest MSE. At the highest MSE end lie the most aggressive GAN- and VAE-based baselines (e.g., CPP-DeID at large λ), indicating large pixel distortions. Toward the lower-error end, diffusion-based methods at mild settings (Mean-, Gauss-, Lpl-GDDiff with small λ) and MaskDDPG at low λ appear, showing that these approaches introduce only slight pixel perturbations. MaskDDPG consistently falls among the group with the least pixel error, especially at its gentlest settings, whereas GDDiFF variants occupy a band of relatively low MSE values before reaching MaskDDPG's region. DeepPrivacy also appears near the low-error region, reflecting that it preserves pixel-level fidelity better than most diffusion or GAN baselines. Although MSE and FD often correlate—techniques with very low MSE tend to show low FD—the relationship is not perfectly linear. Some methods can have similar pixel errors while differing substantially in FD, indicating that identical pixel-level distortions may produce differ-

Table 6.7: Performance metrics on the LFW dataset, sorted in decreasing order by the Fréchet Distance (FD).

Technique	**FD** ↓	KD ↓	P ↑	R ↑	D ↑	C ↑
CPP-DeID (0.9)	2121.45	11.66	0.91	0.00	0.61	0.04
CPP-DeID (0.5)	1531.82	7.00	0.83	0.01	0.47	0.11
CPP-DeID (0.2)	1380.80	6.00	0.72	0.02	0.32	0.11
CPP-DeID (0.0)	1335.30	5.66	0.68	0.02	0.31	0.12
AMT-GAN	820.16	4.32	0.70	0.23	0.29	0.49
FAMS	804.29	2.84	0.67	0.60	0.26	0.43
LDFA	571.87	1.92	0.64	0.80	0.24	0.62
Mean-GDDiff (0.5)	362.03	1.16	0.88	0.80	0.66	0.74
FADM	332.70	0.97	0.82	0.91	0.50	0.82
Mean-GDDiff (0.2)	311.48	1.00	0.91	0.86	0.71	0.81
CLEANIR	303.70	1.24	0.93	0.98	0.46	0.93
Mean-GDDiff (0.1)	289.30	0.93	0.92	0.89	0.73	0.85
Gauss-GDDiff	263.66	0.87	0.94	0.92	0.75	0.89
Mean-GDDiff (0.0)	255.43	0.84	0.94	0.93	0.77	0.90
Lpl-GDDiff	216.63	0.72	0.96	0.96	0.81	0.95
MaskDDPG (1.0)	209.32	0.65	0.94	0.94	0.81	0.93
MaskDDPG (0.8)	201.32	0.63	0.95	0.95	0.82	0.93
MaskDDPG (0.5)	160.36	0.51	0.98	0.98	0.88	0.98
MaskDDPG (0.1)	133.90	0.43	0.99	0.99	0.91	0.99
DeepPrivacy	111.86	0.29	0.99	0.99	0.86	0.98

Table 6.8: Performance metrics on the CALFW dataset, sorted in decreasing order by the Fréchet Distance (FD).

Technique	**FD** ↓	KD ↓	P ↑	R ↑	D ↑	C ↑
CPP-DeID (0.9)	1615.58	8.93	0.90	0.00	0.71	0.07
CPP-DeID (0.5)	1057.98	5.06	0.77	0.02	0.44	0.20
CPP-DeID (0.2)	939.68	4.32	0.71	0.08	0.34	0.22
CPP-DeID (0.0)	893.49	3.98	0.68	0.08	0.33	0.24
AMT-GAN	710.13	3.58	0.80	0.22	0.36	0.51
LDFA	568.54	1.84	0.56	0.68	0.20	0.51
FAMS	548.03	1.93	0.66	0.77	0.28	0.55
CLEANIR	449.87	2.05	0.72	0.82	0.25	0.74
Mean-GDDiff (0.5)	250.10	0.73	0.85	0.79	0.61	0.77
FADM	230.59	0.67	0.85	0.89	0.61	0.87
Mean-GDDiff (0.2)	221.91	0.66	0.89	0.83	0.67	0.84
Mean-GDDiff (0.1)	210.54	0.63	0.91	0.85	0.71	0.86
Gauss-GDDiff	194.99	0.59	0.92	0.87	0.72	0.89
Mean-GDDiff (0.0)	192.51	0.59	0.92	0.88	0.73	0.90
Lpl-GDDiff	168.56	0.52	0.93	0.91	0.76	0.93
MaskDDPG (1.0)	160.41	0.46	0.91	0.90	0.72	0.91
MaskDDPG (0.8)	153.74	0.45	0.92	0.90	0.75	0.92
MaskDDPG (0.5)	130.18	0.39	0.95	0.94	0.83	0.95
MaskDDPG (0.1)	116.10	0.35	0.96	0.95	0.84	0.96
DeepPrivacy	95.29	0.22	0.96	0.96	0.86	0.96

ent perceptual shifts in embedding space. Taken together, FD (which tracks how "far" images move in a learned embedding) and MSE (which tracks raw pixel differences) offer complementary views: FD highlights changes in high-level facial semantics—especially critical for in-the-wild data—while MSE confirms whether low-level fidelity is preserved.

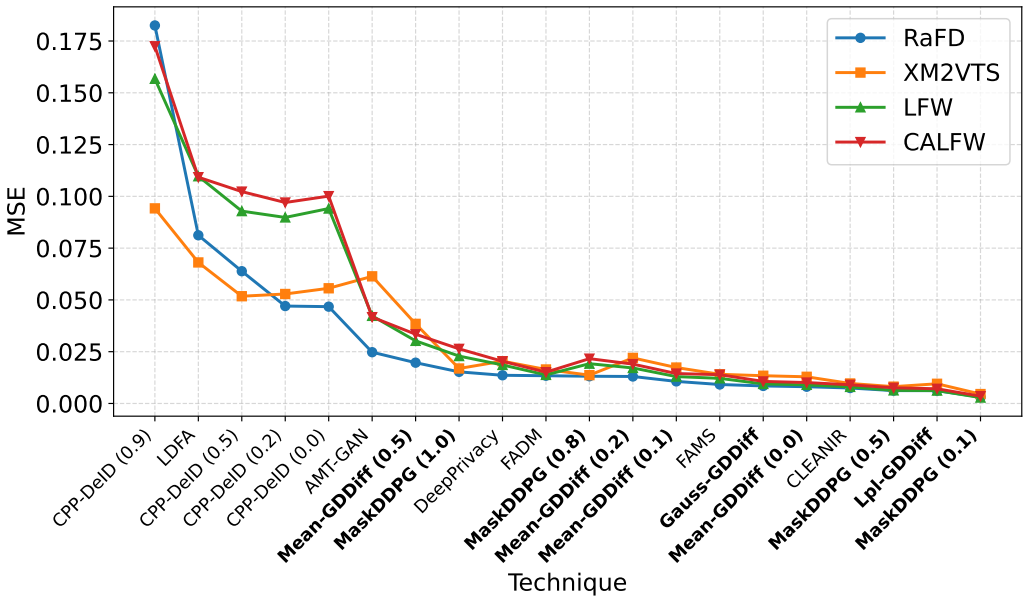


Figure 6.6: Mean MSE per dataset and technique

In short, Figure 6.6 visually confirms that techniques with lower FD (and lower MSE) tend to preserve identity (higher F1), whereas those with higher FD (and higher pixel error) achieve more robust deidentification. Balancing these metrics is therefore dataset-dependent: in clutter-free, studio data FD dominates as the realism increases, while in wild settings density and coverage (alongside FD) ensure that context is maintained. Notably, MaskDDPG consistently remains below the strongest CPP-DeID settings in both FD and MSE while still achieving competitive F1, indicating that our method removes identity effectively without sacrificing undue data quality.

6.3.3 Data Utility Results

Figure 6.7 shows the trade-off between verification AUC (privacy protection) and expression-classification accuracy (HSEmotion) on RaFD. Among the strongest deidentifiers (those with the lowest AUC), MaskDDPG maintains substantially higher expression accuracy than competing methods. For example, at $\lambda = 1.0$, MaskDDPG achieves an AUC around 0.84 while still preserving roughly 40% expression accuracy—nearly double the performance of DeepPrivacy (which yields expression accuracy below 20%). Even at $\lambda = 0.8$, MaskDDPG sits near AUC≈0.86 with expression accuracy ≈40%, whereas CLEANIR in a similar but better AUC; slightly better accuracy ($\geq$ 40%). For the case of GDDiff variants, all of them perform in the range of 20% to 33% approximately showing the sacrifice of using kernels directly in the facial region which erases this important features that express emotions. LDFA which was a clear competitor in AUC metrics, it achieved bad results in expression accuracy with ≈10%. DeepPrivacy was also a clear competitor in verification results and data quality, but here in expression accuracy does poorly with less than 20% of expression accuracy. Deidentifiers that did above 0.86 in AUC tend to retain more expression accuracy. MaskDDPG consistently retains higher expression-classification accuracy, demonstrating that it better preserves facial expression cues even under strong deidentification.

Figure 6.8 depicts gender-classification accuracy (ResNet18) versus verification AUC. Again, MaskDDPG sits near the front: at $\lambda = 1.0$ (AUC≈0.84), it still achieves gender accuracy around 80%. Similar scenario happens with our approach GDDiff, which confirms that it is a solid approach by keeping consistency in gender features by maintaining results approximately to 80%. This shows that even applying strong deidentification as Mean-GDDiff (0.5), you can still be consistent in gender features. By contrast, CPP-DeID at $\lambda = 0.9$ (AUC≈0.65) drops below 65% gender accuracy, and LDFA or CLEANIR in the 0.80–0.85 AUC range only reach about 65–75%. FAMS retained strong data quality metrics and some verification results; however, this

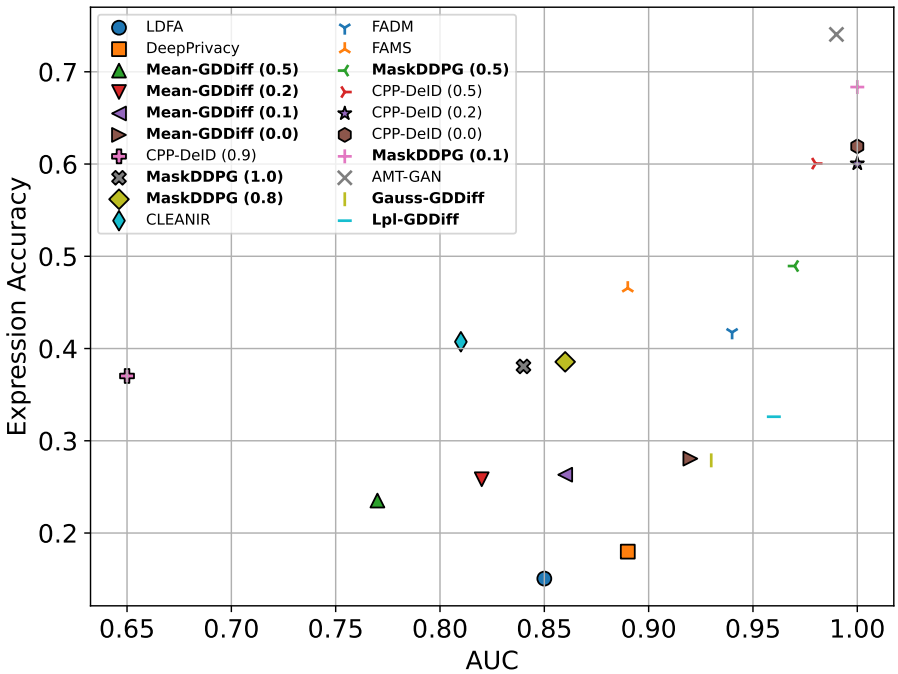


Figure 6.7: Expression accuracy over RaFD dataset using HSEmotion [69] against AUC for privacy protection

method was not able to maintain a good trade-off between identity preservation and gender retention ($\approx$ 40%). The same pattern occurs here: the least effective deidentifiers achieve higher gender accuracy because they make fewer visual changes, thereby preserving the image's natural features.

This experiment shows how our method, MaskDDPG, prioritizes retaining the data-utility trade-off: there is no benefit to deidentification methods achieving near-random AUC if they fail to preserve non-identity features of the subject. This data-utility preservation is crucial, since downstream applications—such as expression recognition, demographic analysis, and other behavioral or cognitive studies—depend on accurately retaining those non-identity attributes. By maintaining these features, MaskDDPG enables the generation of deidentified datasets that remain valuable for research and prac-

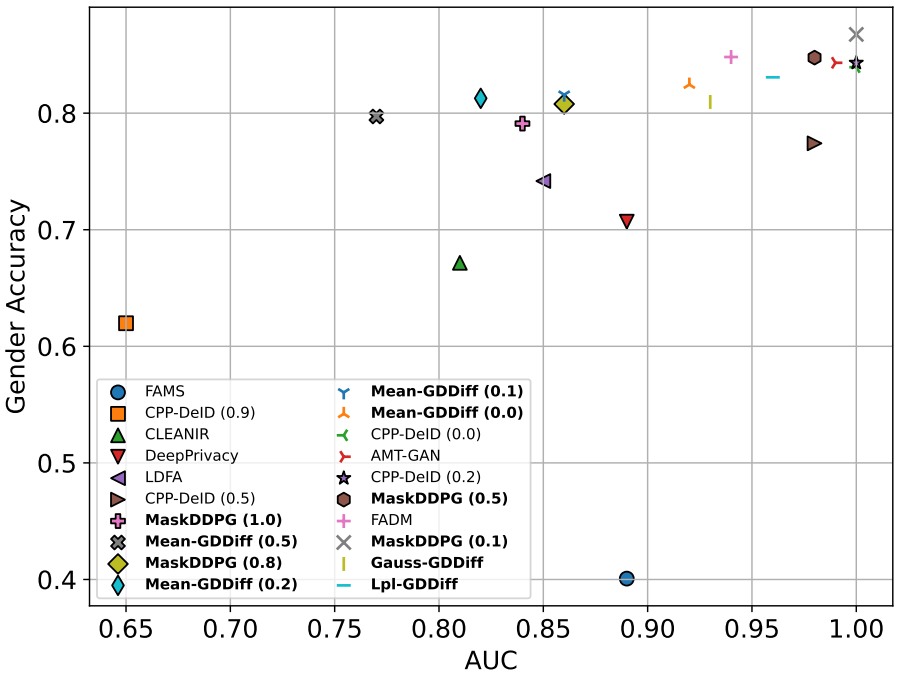


Figure 6.8: Gender accuracy over RaFD dataset using ResNet18 [70] for Gender classification against AUC for privacy protection

tical use without compromising individual privacy. For the case of GDDiff, we partially keep data utility by sacrificing expression features of the face, this is an important aspect that needs to be taken into consideration when using this approach. Even though, it is important to highlight that the identity removal done by GDDiff is stronger than MaskDDPG and the vast majority of the literature review competitors.

6.3.4 Big Picture

In Figure 6.9, we present radar plots summarizing AUC, F1 score, FD, and MSE across all four datasets. Since ground-truth labels for gender and expression are only available for RaFD, the EX and GD results are shown exclusively for that dataset. Each radar plot uses two metrics per "angle" of

analysis: AUC and F1 score for verification performance, FD and MSE for data-quality assessment, and EX and GD accuracies for data-utility evaluation.

These radar charts clearly demonstrate that, at its strongest settings, MaskDDPG and GDDiff enclose a larger area than any competing method. In other words, MaskDDPG and GDDiff achieve robust deidentification (low inverted AUC and F1) while still maintaining high data quality (low FD and MSE) and preserving downstream utility (high EX and GD). In exception of GDDiff that sacrifices expression accuracy. By maximizing the covered area in each radar, we ensure that the deidentified images remain visually realistic and retain important nonidentity features—such as facial expression and gender—despite effectively removing subject identity.

Note that any technique that fails to remove identity will not reach areas above 50% of the maximum radar-plot area, because preserving identity inflates data quality metrics. In contrast, MaskDDPG and GDDiff exceeds 50% of the full radar area at its strongest deidentification setting, indicating a balanced trade-off across all angles of analysis: privacy protection, strong data quality, and good data utility.

By comparison, other methods exhibit clear shortcomings. DeepPrivacy, for instance, often struggles to preserve expression features; FAMS maintains expression well but frequently flips gender labels; CPP-DeID excels at identity removal but shows drastically poor data-quality (realism) metrics; CLEANIR is consistent in identity removal and pixel-level fidelity yet performs poorly in FD; and LDFA lags behind in both expression accuracy and pixel-level error—producing noticeable artifacts in XM2VTS, LFW, and CALFW. Finally, FADM and AMT-GAN have difficulty removing identity altogether, which is reflected by their comparatively small radar areas.

An additional observation from the MaskDDPG plots at varying λ values is that, although data quality and data utility gradually decrease as λ increases, they are never sacrificed entirely (unlike CPP-DeID, which collapses to near zero quality at high privacy levels and GDDiff dropping expression

accuracy). This behavior reinforces that MaskDDPG consistently balances privacy with image realism and utility across all evaluation criteria. Building on these radar-plot insights, it is clear that MaskDDPG and GDDiff design principles—namely, explicitly controlling the privacy-utility trade-off via the λ parameter and focusing transformations on the facial region—yield consistently superior balances across heterogeneous conditions. In practice, this means that our approaches can be fine-tuned for different operational requirements: for example, in a healthcare setting where preserving subtle facial cues is paramount, one could select a moderate λ to maintain high expression accuracy and downstream utility while still ensuring robust identity concealment. Conversely, in surveillance applications where identity protection must be near-absolute (even at the cost of some visual quality) the higher λ settings achieve near-random verification performance with only incremental hits to data quality metrics (FD and MSE values that remain well below those of comparable baselines). This flexibility underscores MaskDDPG's practical viability in real-world pipelines, especially when paired with plug-and-play verification or analytics modules that expect consistent embedding distributions.

This comprehensive "big-picture" analysis shows that our approaches give a novel point in the deidentification landscape: one where neither identity removal nor data fidelity is heavily sacrificed. By systematically optimizing across four distinct datasets, two verification backbones (studio and wild), and three downstream tasks (expression, gender, and radar-based holistic scoring), we demonstrate that MaskDDPG and GDDiff can be reliably adapted to a wide range of operational contexts.

6.3.5 Results

Figure 6.10 shows that the visual outputs align exactly with the trends we observed earlier in Figure 6.9. This confirms the importance of presenting our analysis from multiple perspectives, as highlighted in the previous quantitative experiments.

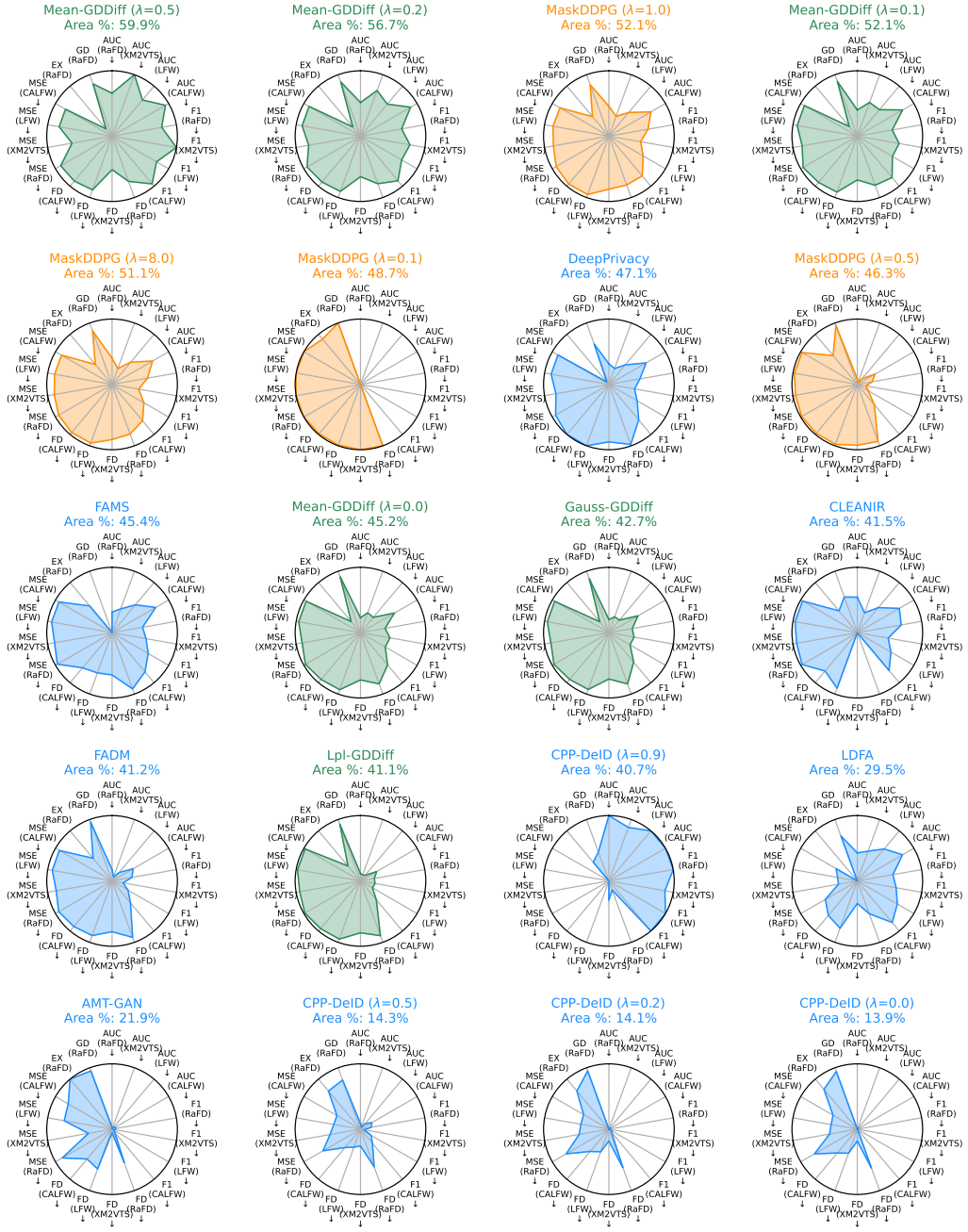


Figure 6.9: Radar plot of key metrics (AUC, F1, FD, MSE, EX, GD) comparing GauDeiDiff (GDDiff - green) as well as MaskDDPG (blue) and state-of-the-art approaches on RaFD, XM2VTS, LFW, and CALFW. Lower AUC, F1, FD, and MSE values imply better performance (larger plot areas).

Figure 6.10 also shows that GDDiff often fails to replicate facial expression details and even removes glasses from subjects who wore them in the original images. In contrast, MaskDDPG avoids these issues by producing slightly less-obfuscated outputs. Moreover, MaskDDPG preserves the background as much as possible, resulting in higher-quality images when the background is variable (e.g., in LFW). Regardless of which method is used, both MaskDDPG and GDDiff preserve skin color, pose, and gender—features that are essential for maintaining consistency with the original samples.

Moreover, Figure 6.11 shows the outputs generated by all competing methods. CPP-DeID fails to generate realistic-looking images, producing altered backgrounds and significant inconsistencies in gender and expression. DeepPrivacy, on the other hand, generates realistic images but does not preserve the subjects' apparent ages—most outputs resemble children. FAMS is inconsistent in terms of gender: it often swaps the subject's gender in the generated images. CLEANIR fails to produce visually appealing outputs and also struggles to maintain gender consistency. FADM, a diffusion-based approach, generates high-quality images but does not sufficiently remove identity. AMT-GAN and LDFA struggle to generate high-quality images. MaskDDPG and GDDiff are consistent in several aspects, as shown in these visual results: they effectively remove identity from facial images while preserving data quality and utility. These visual results reinforce the claims made in the previous subsections.

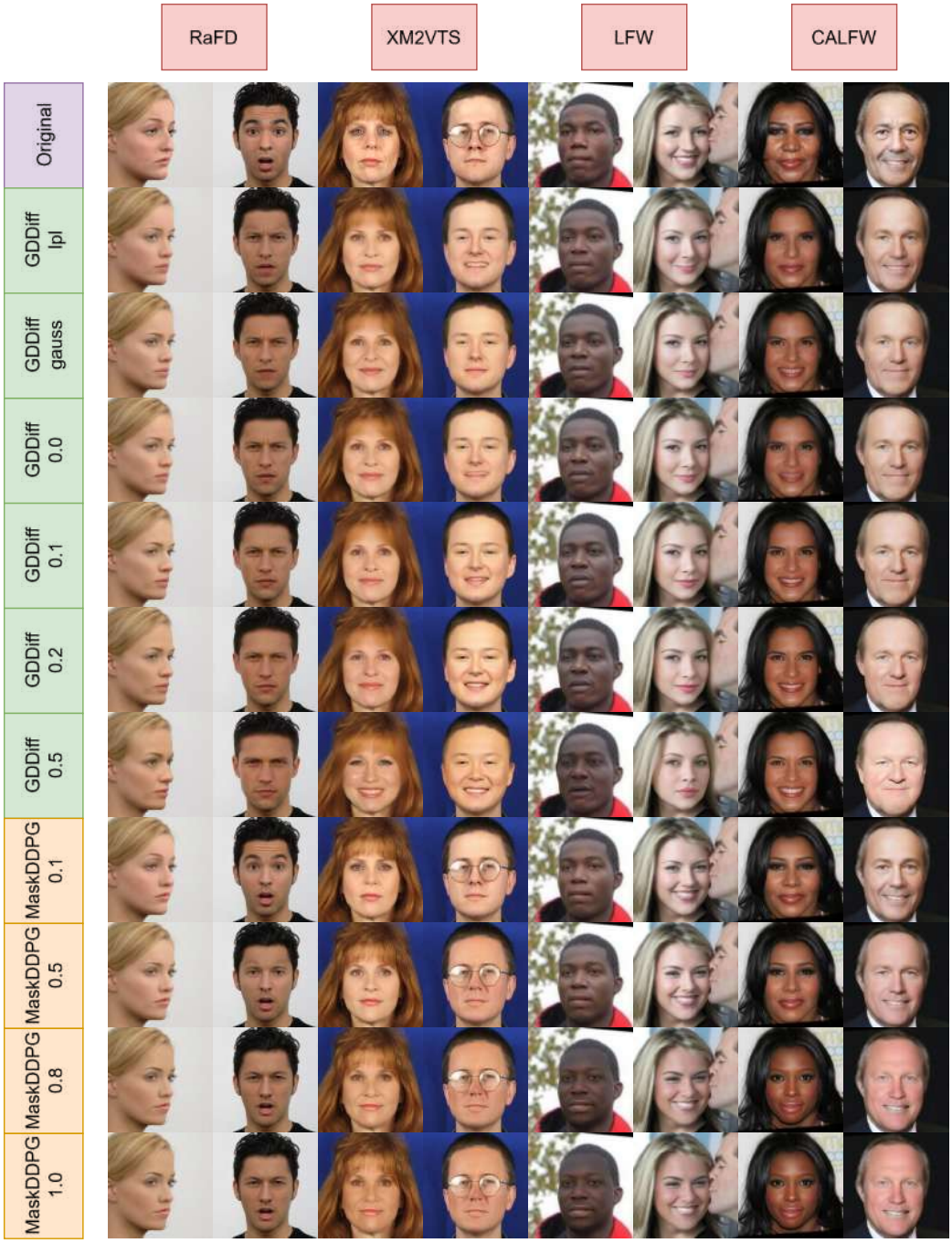


Figure 6.10: Visual results for MaskDDPG and GauDeiDiff (GDDiff)

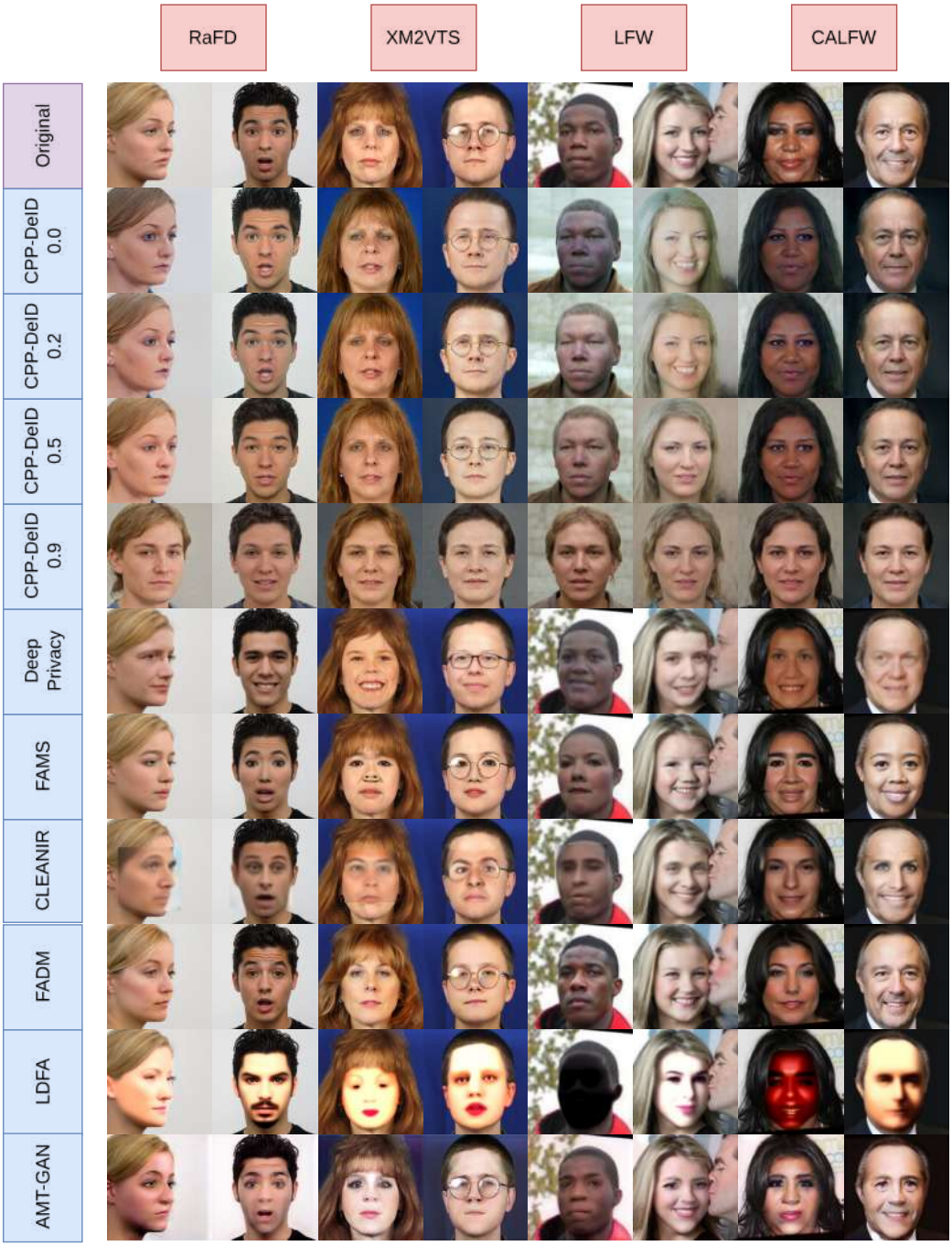
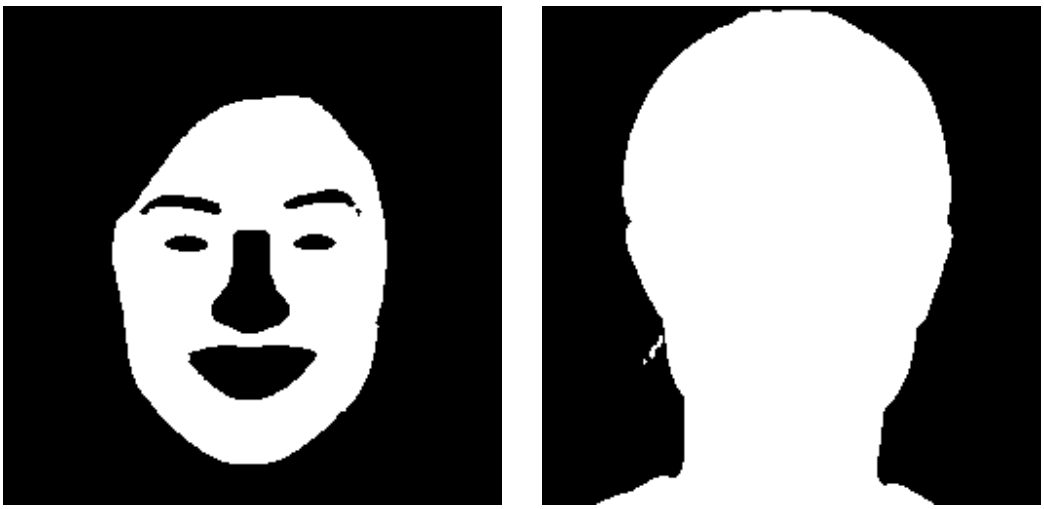


Figure 6.11: Competitor results

6.3.6 Ablation Studies

Ablation Studies 1 & 2: mask selection

For this section, we have two main ablation studies. In the first ablation study, the binary mask indicates that our mechanism is applied only to the skin region of the face (see Figure 6.12a). In the second ablation study, we cover the subject's entire silhouette and apply our mechanism over that region.



(a) Ablation study 1 (Abs1): de-identification applied only to the facial skin area based on the selected mask.

(b) Ablation study 2 (Abs2): de-identification applied to the entire silhouette of the person.

Figure 6.12: Example of the binary masks used in these ablation studies. White pixels indicate where our mechanism is going to be applied.

For these ablation studies, Figure 6.13 shows how the first ablation study (Abs1) prioritizes data quality and data utility by achieving high gender accuracy (GD), high expression accuracy, high FD, and high MSE on the RaFD dataset, which was the only dataset with gender and expression labels available for benchmarking. On the other hand, the second ablation study (Abs2) sacrifices data quality and utility as λ increases in order to prioritize verification (i.e., inverted metrics of AUC and F1 score). These ablation

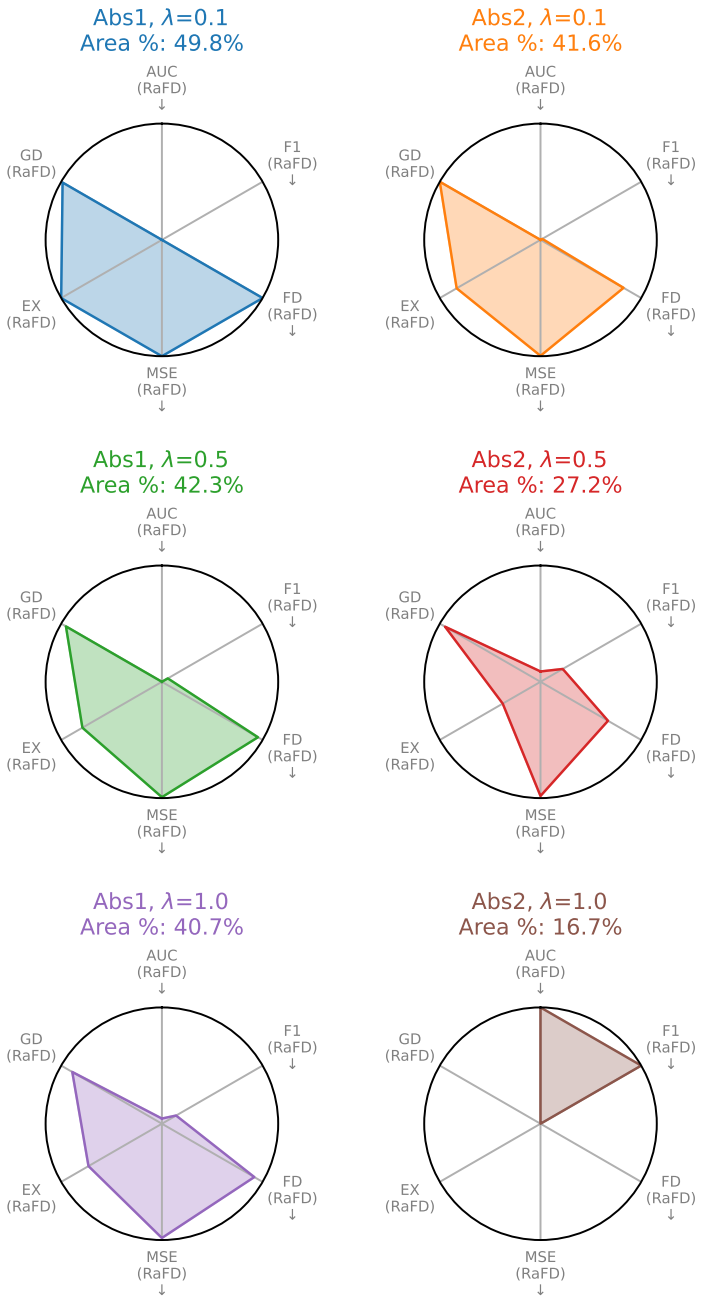


Figure 6.13: Summary of the main metrics over the ablation studies on RaFD dataset

studies demonstrate that our original binary-mask setting (facial skin, both brows, both eyes, and the nose receive our deidentification method) achieves the desired performance balance between ID removal, data quality, and data utility across the evaluated datasets, as shown previously in Figure 6.9.

Ablation study 3: influence of glasses in deidentification

In this experiment, we used GlassesGAN [71] which is an image editing framework for custom design of glasses. This framework was used to add transparent glasses to all images in the RaFD dataset. The goal was to measure how much these glasses affect verification performance. This is important because our deidentification method does not modify the region occupied by glasses when they are present. We selected RaFD because it is a studio dataset with minimal variability from external factors such as illumination or background. Moreover, since most subjects in RaFD do not originally wear glasses, it is appropriate to use GlassesGAN to insert transparent glasses artificially. We chose transparent glasses so that the eyes and a large portion of the face remain visible, potentially revealing identity cues.

For this ablation study (Abs3), we modify our binary mask to include the glasses region, thereby applying identity removal over the glasses as well. First, we generated "glasses" images using GlassesGAN. Next, we applied deidentification under two settings: (1) our strongest MaskDDPG configuration ($\lambda = 1.0$), which does not remove identity from the glasses region; and (2) the Abs3 configuration, in which the binary mask covers both the face and the glasses. In the visual results shown in Figure 6.14, illustrate MaskDDPG at $\lambda = 1.0$; the glasses remain intact while identity is removed from the underlying face. In the rightmost column, Abs3 removes the glasses entirely (alongside other identity cues), producing noticeable changes in the image and even altering perceived gender in some cases (e.g., the third image).

Figure 6.15 compares the ROC curves for MaskDDPG at $\lambda = 1.0$ (not removing identity from the glasses) and Abs3 (removing identity from the glasses). Removing identity from the glasses region reduces the AUC to

0.82, compared to 0.92 when glasses are left untouched. This indicates that the glasses region contributes approximately 10% of AUC. This means that it carries significant identity information. However, because preserving data utility (maintaining realistic glasses) is a priority, we accept this 10% increase in AUC rather than removing identity from the glasses region.

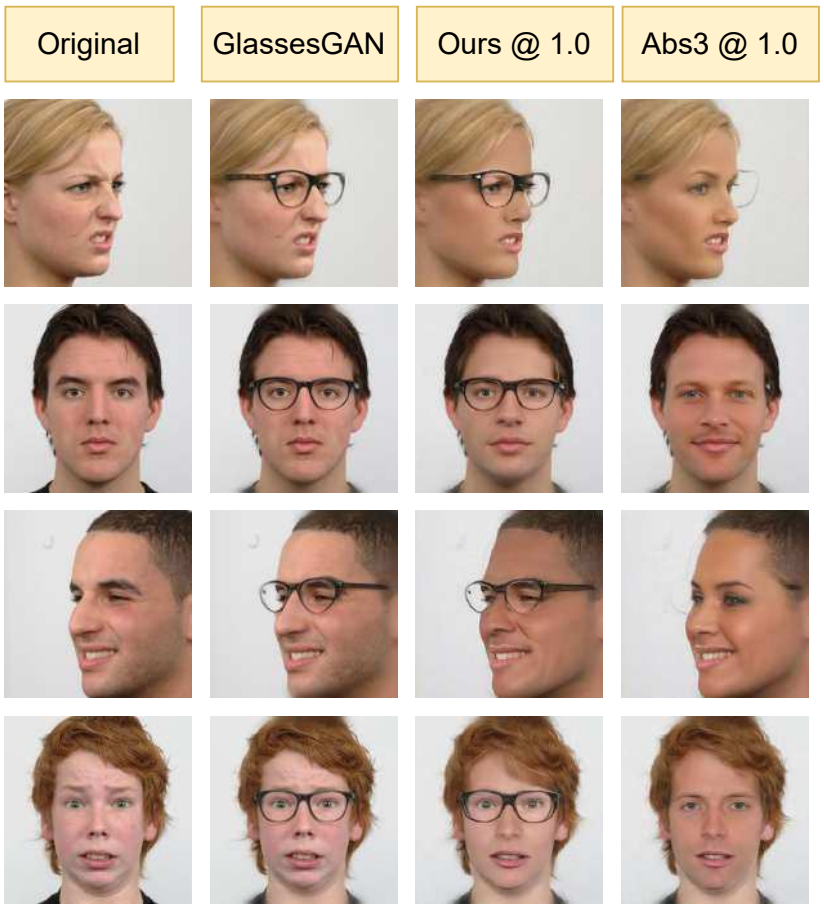


Figure 6.14: Visual results of the ablation study 3 using GlassesGAN over RaFD. Comparision of our vanilla MaskDDPG using $\lambda = 1.0$ and the ablation study 3 (Abs3) setting

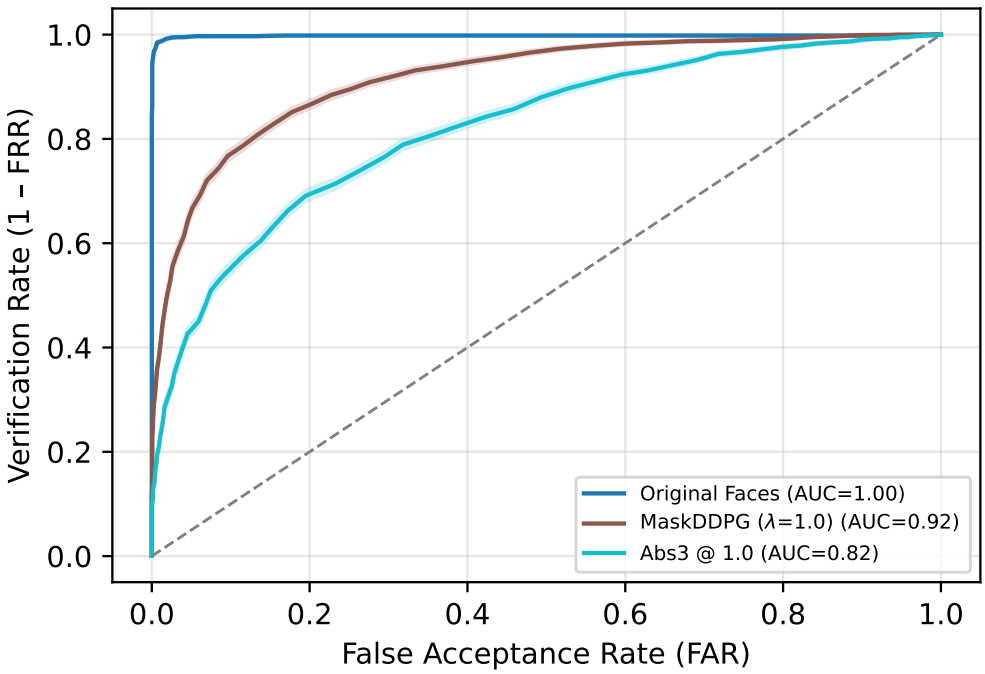


Figure 6.15: ROC curves for the comparison of our approach MaskDDPG using $\lambda = 1.0$ and the ablation study 3 (Abs3) experiment over the RaFD dataset.

Chapter 7

Conclusion

In this work, we have introduced two novel methods such as MaskDDPG and GauDeiDiff for controllable face deidentification within diffusion-model frameworks. The motivation behind our research stemmed from the growing need to protect individual privacy in an era of ubiquitous face imagery, where naive obfuscation techniques such as blurring or pixelation either fail to prevent re-identification or severely degrade image utility. By leveraging the denoising diffusion process with identity-aware guidance, our approaches aim to strike a careful balance between privacy protection, visual realism, and preservation of non-identity attributes.

MaskDDPG builds upon a state-of-the-art diffusion sampler by introducing two key elements: a user-defined privacy parameter λ that controls guidance strength, and a spatial segmentation mask that designates which facial regions to anonymize. This combination allows MaskDDPG to progressively weaken identity similarity while retaining soft-biometric features such as expression, glasses, and background context. Through extensive experiments on both studio datasets (RaFD, XM2VTS) and in-the-wild collections (LFW, CALFW), we demonstrated that MaskDDPG consistently achieves strong deidentification, pushing verification performance toward low levels—while maintaining low Fréchet Distance and mean squared error relative to competing GAN- and VAE-based methods. Moreover, downstream tasks such

as expression and gender classification on RaFD revealed that MaskDDPG preserves non-identity cues more effectively than all baselines, even at its strongest privacy setting.

Complementing MaskDDPG, we proposed GauDeiDiff, a framework that applies classical spatial kernels (Gaussian and Laplacian-of-Gaussian) directly to the guidance tensors without requiring any additional model training. By scheduling an incremental variance for these kernels across denoising steps and optionally injecting an "average-face" inference at a user-specified timestep, GauDeiDiff produces multiple deidentified outputs at distinct privacy levels. In studio settings, this kernel-based approach matched or exceeded MaskDDPG's performance in terms of verification metrics while delivering image fidelity that remained competitive with more complex baselines. The "average-face" branch further allows users to control the randomness of deidentification: when introduced early in the Markov chain, it yields near-random faces; when introduced later, it produces subtler, more realistic alterations. Although GauDeiDiff sacrifices some expression accuracy compared to MaskDDPG—owing to broader kernel-based perturbations—it still preserves gender and pose cues better than many state-of-the-art competitors.

Our ablation studies shed light on the importance of spatial selectivity. Restricting the MaskDDPG guidance to only key facial regions (skin, eyes, brows, nose) achieved an optimal trade-off between privacy, data quality, and utility. Expanding the mask to cover the entire silhouette improved verification performance at the cost of drastically reduced image quality, expression and gender accuracy, while including the glasses region added approximately a 10% gain in AUC but produced visually noticeable artifacts and altered non-identity attributes. These findings confirm that carefully choosing which regions to mask is crucial: masking too broadly undermines utility, whereas masking too narrowly fails to sufficiently obscure identity.

Despite these advances, several limitations remain. MaskDDPG's effectiveness hinges on accurate facial segmentation; in scenarios with extreme

poses, occlusions, or low resolution, mask extraction may falter, leading to uneven deidentification or artifacts. Additionally, our evaluation focused primarily on identity, expression, and gender preservation; other sensitive attributes (age, race, health markers) were not explicitly protected or measured. Finally, we did not provide formal privacy guarantees (such as differential privacy), relying instead on empirical verification metrics (AUC, EER, F1) to quantify identity removal. Incorporating theoretical privacy bounds into the diffusion framework remains an important future direction.

Looking forward, several avenues can further enhance these methods. Formalizing privacy through information-theoretic measures or adversarial objectives may yield stronger guarantees against re-identification. Finally, extending these techniques to video, with temporal consistency constraints, would enable coherent deidentification across frames, avoiding flickering artifacts and ensuring stable anonymization in motion.

In conclusion, MaskDDPG and GauDeiDiff offer practical, flexible, and high-fidelity solutions for face deidentification. By embedding identity-aware guidance into the diffusion process and leveraging spatial control, our methods demonstrate that strong privacy protection need not come at the expense of image realism or utility. The ability to tune the privacy parameter λ and to specify which facial regions to anonymize provides end users with fine-grained control suited to diverse operational requirements—from privacy-critical surveillance to utility-critical research datasets. As diffusion models continue to advance, we anticipate that the principles laid out here; controllable guidance, spatial masking, and kernel-based regularization—will serve as a foundation for next-generation privacy-preserving generative systems, enabling wide access to deidentified visual data without compromising individual rights.

Bibliography

[1] The International Association of Privacy Professionals (IAPP), What is Privacy (2000).
URL https://iapp.org/about/what-is-privacy/

[2] Equality and Human Rights Commission, The Human Rights Act, Article 8: Respect for your private and family life, https://www.equalityhumanrights.com/en/human-rights-act/article-8-respect-your-private-and-family-life (2022).

[3] P. Agrawal, P. J. Narayanan, Person de-identification in videos, IEEE Transactions on Circuits and Systems for Video Technology 21 (3) (2011) 299–310.

[4] E. Newton, L. Sweeney, B. Malin, Preserving privacy by de-identifying face images, IEEE Transactions on Knowledge and Data Engineering 17 (2) (2005) 232–243. doi:10.1109/TKDE.2005.32.

[5] R. Gross, L. Sweeney, J. Cohn, F. de la Torre, S. Baker, Face De-identification, 2009, pp. 129–146. doi:10.1007/978-1-84882-301-3_8.

[6] B. Meden, P. Rot, P. Terhörst, N. Damer, A. Kuijper, W. J. Scheirer, A. Ross, P. Peer, V. Štruc, Privacy–enhancing face biometrics: A comprehensive survey, IEEE Transactions on Information Forensics and Security 16 (2021) 4147–4183. doi:10.1109/TIFS.2021.3096024.

[7] K. Kobayashi, K. Iwamura, K. Kaneda, I. Echizen, Surveillance camera system to achieve privacy protection and crime prevention, in: IIH-MSP, 2014, pp. 463–466. doi:10.1109/IIH-MSP.2014.122.

[8] A. Das, M. Degeling, X. Wang, J. Wang, N. Sadeh, M. Satyanarayanan, Assisting users in a world full of cameras: A privacy-aware infrastructure for computer vision applications, in: CVPR Workshops, 2017, pp. 1387–1396. doi:10.1109/CVPRW.2017.181.

[9] N. A. Van House, Collocated photo sharing, story-telling, and the performance of self, International Journal of Human-Computer Studies 67 (12) (2009) 1073–1086. doi:https://doi.org/10.1016/j.ijhcs.2009.09.003.

[10] B. Meden, Z. Emersic, V. Struc, P. Peer, κ-same-net: Neural-network-based face deidentification, in: International Work Conference on Bioinspired Intelligence (IWOBI), 2017, pp. 1–7.

[11] B. Meden, R. C. Mallı, S. Fabijan, H. K. Ekenel, V. Štruc, P. Peer, Face deidentification with generative deep neural networks, IET Signal Processing 11 (9) (2017) 1046–1054.

[12] B. Meden, Ž. Emeršič, V. Štruc, P. Peer, k-same-net: k-anonymity with generative deep neural networks for face deidentification, Entropy 20 (1) (2018) 60.

[13] W. L. Croft, J.-R. Sack, W. Shi, Differentially private obfuscation of facial images, in: Machine Learning and Knowledge Extraction, 2019, pp. 229–249.

[14] V. Mirjalili, S. Raschka, A. Ross, Gender Privacy: An Ensemble of Semi Adversarial Networks for Confounding Arbitrary Gender Classifiers, in: 2018 IEEE 9th International Conference on Biometrics Theory, Applications and Systems (BTAS), IEEE, Redondo Beach, CA, USA, 2018, pp. 1–10. doi:10.1109/BTAS.2018.8698605.

[15] L. Yuan, L. Liu, X. Pu, Z. Li, H. Li, X. Gao, Pro-face: A generic framework for privacy-preserving recognizable obfuscation of face images, in: Proceedings of the 30th ACM International Conference on Multimedia, MM '22, Association for Computing Machinery, New York, NY, USA, 2022, p. 1661–1669. doi:10.1145/3503161.3548202.

[16] D. Cho, J. H. Lee, I. H. Suh, CLEANIR: Controllable Attribute-Preserving Natural Identity Remover, Applied Sciences 10 (3) (2020) 1120. doi:10.3390/app10031120.

[17] M. Maximov, I. Elezi, L. Leal-Taixe, CIAGAN: Conditional Identity Anonymization Generative Adversarial Networks, in: CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, Seattle, WA, USA, 2020, pp. 5446–5455. doi:10.1109/CVPR42600.2020.00549.

[18] S. M. Sadegh Moosavi Khorzooghi, S. Nilizadeh, Stylegan as a utility-preserving face de-identification method, arXiv preprint (2022) arXiv–2212.

[19] H. Hukkelås, R. Mester, F. Lindseth, Deepprivacy: A generative adversarial network for face anonymization, in: ISVC, 2019, p. 565–578. doi:10.1007/978-3-030-33720-9_44.

[20] H. Hukkelås, F. Lindseth, Deepprivacy2: Towards realistic full-body anonymization, in: WACV, 2023, pp. 1329–1338.

[21] S. Hu, X. Liu, Y. Zhang, M. Li, L. Y. Zhang, H. Jin, L. Wu, Protecting facial privacy: Generating adversarial identity masks via style-robust makeup transfer, in: CVPR, 2022, pp. 15014–15023.

[22] Y. Peng, A Comparative Analysis Between GAN and Diffusion Models in Image Generation, Transactions on Computer Science and Intelligent Systems Research 5 (2024) 189–195. doi:10.62051/0f1va465.

[23] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-Resolution Image Synthesis with Latent Diffusion Models, in: CVPR, 2022, pp. 10674–10685. doi:10.1109/CVPR52688.2022.01042.

[24] L. Zhang, A. Rao, M. Agrawala, Adding Conditional Control to Text-to-Image Diffusion Models, in: 2023 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE, Paris, France, 2023, pp. 3813–3824. doi:10.1109/ICCV51070.2023.00355.

[25] T. Tirer, Iteratively preconditioned guidance of denoising (diffusion) models for image restoration, in: ICASSP, 2024, pp. 2465–2469. doi:10.1109/ICASSP48485.2024.10446692.

[26] T. Tirer, R. Giryes, Back-projection based fidelity term for ill-posed linear inverse problems, IEEE Transactions on Image Processing 29 (2020) 6164–6179. doi:10.1109/TIP.2020.2988779.

[27] E. M. Bender, T. Gebru, A. McMillan-Major, S. Shmitchell, On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?, in: Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency, ACM, Virtual Event Canada, 2021, pp. 610–623. doi:10.1145/3442188.3445922.

[28] M. Kim, A. K. Jain, X. Liu, AdaFace: Quality Adaptive Margin for Face Recognition, in: CVPR, 2022, pp. 18729–18738. doi:10.1109/CVPR52688.2022.01819.

[29] M. Gong, J. Liu, H. Li, Y. Xie, Z. Tang, Disentangled representation learning for multiple attributes preserving face deidentification, IEEE Transactions on Neural Networks and Learning Systems 33 (1) (2022) 244–256. doi:10.1109/TNNLS.2020.3027617.

[30] H. Proença, The uu-net: Reversible face de-identification for visual surveillance video footage, IEEE Transactions on Circuits and Systems

for Video Technology 32 (2) (2022) 496–509. doi:10.1109/TCSVT.2021.3066054.

[31] B. Meden, M. Gonzalez-Hernandez, P. Peer, V. Štruc, Face deidentification with controllable privacy protection, IMAVIS 134 (2023) 104678. doi:https://doi.org/10.1016/j.imavis.2023.104678.

[32] D. Li, W. Wang, K. Zhao, J. Dong, T. Tan, Riddle: Reversible and diversified de-identification with latent encryptor, arxiv preprint abs/2303.05171 (2023).

[33] J. Song, Y. Jin, Y. Li, C. Lang, Learning structural similarity with evolutionary-gan: A new face de-identification method, in: 2019 6th International Conference on Behavioral, Economic and Socio-Cultural Computing (BESC), 2019, pp. 1–6. doi:10.1109/BESC48373.2019.8962993.

[34] Y. Qiu, Z. Niu, B. Song, T. Ma, A. Al-Dhelaan, M. Al-Dhelaan, A novel generative model for face privacy protection in video surveillance with utility maintenance, Applied Sciences 12 (14) (2022). doi:10.3390/app12146962.

[35] Z. Ren, Y. J. Lee, M. S. Ryoo, Learning to anonymize faces for privacy preserving action detection, in: Proceedings of the European Conference on Computer Vision (ECCV), 2018.

[36] H. Hukkelås, R. Mester, F. Lindseth, Deepprivacy: A generative adversarial network for face anonymization, in: Advances in Visual Computing: 14th International Symposium on Visual Computing, ISVC 2019, Lake Tahoe, NV, USA, October 7–9, 2019, Proceedings, Part I, Springer-Verlag, 2019, p. 565–578. doi:10.1007/978-3-030-33720-9_44.

[37] Z. Huang, K. C. Chan, Y. Jiang, Z. Liu, Collaborative Diffusion for Multi-Modal Face Generation and Editing, in: CVPR, 2023, pp. 6080–6090. doi:10.1109/CVPR52729.2023.00589.

[38] M. Kim, F. Liu, A. Jain, X. Liu, Dcface: Synthetic face generation with dual condition diffusion model, in: CVPR, 2023, pp. 12715–12725. doi:10.1109/CVPR52729.2023.01223.

[39] B. T. Morvay, B. Béres, S. Torma, L. Szegletes, Diffusion Probabilistic Model Based Face Anonymization in Embedded Environments, in: CogInfoCom, 2023, pp. 000135–000140. doi:10.1109/CogInfoCom59411.2023.10397492.

[40] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, in: H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin (Eds.), Advances in Neural Information Processing Systems, Vol. 33, 2020, pp. 6840–6851.

[41] L. Piano, P. Basci, F. Lamberti, L. Morra, Latent diffusion models for attribute-preserving image anonymization (2024).

[42] H. Ye, J. Zhang, S. Liu, X. Han, W. Yang, Ip-adapter: Text compatible image prompt adapter for text-to-image diffusion models (2023). arXiv:2308.06721.
URL https://arxiv.org/abs/2308.06721

[43] C. Mou, X. Wang, L. Xie, Y. Wu, J. Zhang, Z. Qi, Y. Shan, T2i-adapter: learning adapters to dig out more controllable ability for text-to-image diffusion models, in: Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence, AAAI'24/IAAI'24/EAAI'24, AAAI Press, 2024. doi:10.1609/aaai.v38i5.28226.

[44] M. Klemp, K. Rösch, R. Wagner, J. Quehl, M. Lauer, LDFA: Latent Diffusion Face Anonymization for Self-driving Applications, in: CVPR Workshops, 2023, pp. 3199–3205. doi:10.1109/CVPRW59228.2023.00322.

[45] X. He, M. Zhu, D. Chen, N. Wang, X. Gao, Diff-privacy: Diffusion-based face privacy protection, IEEE Transactions on Circuits and Systems for Video Technology (2024) 1–1doi:10.1109/TCSVT.2024.3449290.

[46] X. You, X. Zhao, Y. Wang, W. Sun, Generation of face privacy-protected images based on the diffusion model, Entropy 26 (6) (2024). doi:10.3390/e26060479.

[47] H.-W. Kung, T. Varanka, S. Saha, T. Sim, N. Sebe, Face Anonymization Made Simple , in: 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), IEEE Computer Society, Los Alamitos, CA, USA, 2025, pp. 1040–1050. doi:10.1109/WACV61041.2025.00110.

[48] H. Li, Animate Anyone: Consistent and Controllable Image-to-Video Synthesis for Character Animation , in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE Computer Society, Los Alamitos, CA, USA, 2024, pp. 8153–8163. doi:10.1109/CVPR52733.2024.00779.

[49] P. Zwick, K. Roesch, M. Klemp, O. Bringmann, Context-aware full body anonymization using text-to-image diffusion models, arXiv preprint arXiv:2410.08551 (2024).

[50] M. Shaheryar, J. T. Lee, S. K. Jung, Iddiffuse: Dual-conditional diffusion model for enhanced facial image anonymization, in: Proceedings of the Asian Conference on Computer Vision, 2024, pp. 4017–4033.

[51] H. Uchida, N. Abe, S. Yamada, Dedim: De-identification using a diffusion model, in: International Conference of the Biometrics Special Interest Group (BIOSIG), IEEE, 2022, pp. 1–5.

[52] Y. Zhu, J. Cao, B. Liu, T. Chen, R. Xie, L. Song, Identity-consistent video de-identification via diffusion autoencoders, in: IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB), IEEE, 2024, pp. 1–6.

[53] J. Song, C. Meng, S. Ermon, Denoising diffusion implicit models, in: International Conference on Learning Representations, 2021.
URL https://openreview.net/forum?id=St1giarCHLP

[54] T. Tirer, R. Giryes, Image restoration by iterative denoising and backward projections, IEEE Transactions on Image Processing 28 (3) (2019) 1220–1234. doi:10.1109/TIP.2018.2875569.

[55] J. Song, A. Vahdat, M. Mardani, J. Kautz, Pseudoinverse-guided diffusion models for inverse problems, in: International Conference on Learning Representations, 2023.

[56] T. Tirer, R. Giryes, Super-resolution via image-adapted denoising cnns: Incorporating external and internal learning, IEEE Signal Processing Letters 26 (7) (2019) 1080–1084. doi:10.1109/LSP.2019.2920250.

[57] A. Graikos, N. Malkin, N. Jojic, D. Samaras, Diffusion models as plug-and-play priors, in: Proceedings of the 36th International Conference on Neural Information Processing Systems, Curran Associates Inc., Red Hook, NY, USA, 2022.

[58] Y. Romano, M. Elad, P. Milanfar, The little engine that could: Regularization by denoising (red), SIAM Journal on Imaging Sciences 10 (4) (2017) 1804–1844. doi:10.1137/16M1102884.

[59] T. Tirer, R. Giryes, On the convergence rate of projected gradient descent for a back-projection based objective, SIAM Journal on Imaging Sciences 14 (4) (2021) 1504–1531. doi:10.1137/21M1407902.

[60] J. Dan, Y. Liu, H. Xie, J. Deng, H. Xie, X. Xie, B. Sun, Transface: Calibrating transformer training for face recognition from a data-centric perspective, in: IEEE/CVF International Conference on Computer Vision (ICCV), 2023, pp. 20585–20596.

[61] K. Messer, J. Kittler, M. Sadeghi, S. Marcel, C. Marcel, S. Bengio, F. Cardinaux, C. Sanderson, J. Czyz, L. Vandendorpe, S. Srisuk,

M. Petrou, W. Kurutach, A. Kadyrov, R. Paredes, B. Kepenekci, F. B. Tek, G. B. Akar, F. Deravi, N. Mavity, Face verification competition on the xm2vts database, in: Proceedings of the 4th International Conference on Audio- and Video-Based Biometric Person Authentication, AVBPA'03, Springer-Verlag, 2003, p. 964–974.

[62] O. Langner, R. Dotsch, G. Bijlstra, D. H. Wigboldus, S. T. Hawk, A. van Knippenberg, Presentation and validation of the radboud faces database, Cognition and Emotion 24 (8) (2010) 1377–1388. `doi:10.1080/02699930903485076`.

[63] G. B. Huang, M. Mattar, T. Berg, E. Learned-Miller, Labeled faces in the wild: A database forstudying face recognition in unconstrained environments, in: Workshop on faces in'Real-Life'Images: detection, alignment, and recognition, 2008.

[64] T. Zheng, W. Deng, J. Hu, Cross-age lfw: A database for studying cross-age face recognition in unconstrained environments (2017). `arXiv:1708.08197`.

[65] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, M. Assran, N. Ballas, W. Galuba, R. Howes, P.-Y. Huang, S.-W. Li, I. Misra, M. Rabbat, V. Sharma, G. Synnaeve, H. Xu, H. Jegou, J. Mairal, P. Labatut, A. Joulin, P. Bojanowski, Dinov2: Learning robust visual features without supervision (2024). `arXiv:2304.07193`.

[66] M. S. Sajjadi, O. Bachem, M. Lucic, O. Bousquet, S. Gelly, Assessing generative models via precision and recall, Advances in neural information processing systems 31 (2018).

[67] M. F. Naeem, S. J. Oh, Y. Uh, Y. Choi, J. Yoo, Reliable fidelity and diversity metrics for generative models, in: International Conference on Machine Learning, PMLR, 2020, pp. 7176–7185.

[68] G. Stein, J. Cresswell, R. Hosseinzadeh, Y. Sui, B. Ross, V. Villecroze, Z. Liu, A. L. Caterini, E. Taylor, G. Loaiza-Ganem, Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models, Advances in Neural Information Processing Systems 36 (2024).

[69] A. V. Savchenko, Hsemotion: High-speed emotion recognition library, Software Impacts 14 (2022) 100433.

[70] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778. doi:10.1109/CVPR.2016.90.

[71] R. Plesh, P. Peer, V. Štruc, Glassesgan: Eyewear personalization using synthetic appearance discovery and targeted subspace modeling, in: Proceedings of the 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 16847–16857.